

The Modern Era: McVay, Mahomes, Two-High, and the Under-Center Comeback (2017-2025)

Nine seasons of move and counter-move: the play-action revival, Mahomes, the safeties who stayed deep, the offenses that got bigger to beat them, the quarterback who runs, the coaches who go for it, and the rules that shifted under all of it.

February 8, 2026, Super Bowl LX at Levi's Stadium. The Seattle Seahawks beat the New England Patriots 29-13, and their running back, Kenneth Walker III, carried 27 times for 135 yards and was named the game's MVP. Look at how Seattle got there and the season reads like a list of everything the NFL had supposedly left behind. On offense, the Seahawks took the snap with the quarterback **under center**, right behind the center, on more than half their plays, second-most in the league, and played with fewer than three wide receivers on more than half of them. On defense, they kept two safeties deep on about half of opponents' dropbacks and still sacked Drake Maye six times in the Super Bowl.¹ Nine seasons earlier, the most copied offense in football had three wide receivers on the field for almost every snap, and the defenses it faced usually had one deep safety and eight players crowded near the line. Nearly every piece of that 2017 picture had flipped by 2025. None of it was nostalgia. Each change was an answer to the change before it, and this chapter follows the chain, from Sean McVay's first season in Los Angeles to the morning after Seattle's title, and shows you how to see each link on a Sunday.

You'll be able to...

- Explain what Sean McVay and Kyle Shanahan built in 2017 (one personnel group, tight splits, motion, and play-action off the outside zone) and why it tore up the single-high defenses of the day.
- Explain what Patrick Mahomes and Andy Reid added: a passing game borrowed from college and a quarterback whose best plays happened after the design broke down.
- Explain the **two-high counterrevolution**, read its trend line, and say what it cost defenses; then explain the offensive answer (the **light-box run game**, heavier personnel, and the **under-center revival**) by counting blockers and defenders on a diagram.
- Name the positions that changed shape (slot corner, star safety, hybrid linebacker, move tight end, edge rusher, fullback) and say which move in the arms race each one answered.
- Explain why the quarterback run became normal this time when it faded after 2012, and why coaches started going for it on fourth down, with the numbers behind both.
- List the rule changes of 2017-2026 that matter for strategy and describe where the game stands entering the 2026 season.

¹Super Bowl LX (February 8, 2026, Levi's Stadium, Santa Clara): Seahawks 29, Patriots 13; Kenneth Walker III MVP with 27 carries for 135 yards; Seattle sacked Drake Maye six times; head coach Mike Macdonald, offensive coordinator Klint Kubiak, defensive coordinator Aden Durde. "Super Bowl LX," Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_LX ; Fox 13 Seattle live updates: <https://www.fox13seattle.com/news/super-bowl-live-updates-seahawks-patriots> (both as checked in the course fact sheet). Seattle's 2025 rates are this book's calculation for the regular season: under center on 53.4 percent of runs and passes (FTN charting qb_location, second to the Rams' 59.1), fewer than three wide receivers on 55.2 percent (nflverse participation), and a two-high coverage (Cover 2, 2-Man, Cover 4, Cover 6) on 51.7 percent of opponents' dropbacks with a charted coverage, third behind Minnesota and Buffalo.

i You'll need

This chapter finishes the history of Part 11 and pulls threads from all over the book. One line each, with a link if you want to catch up:

- **The previous era** (Dynasties and Counterpunches): the move tight end forced defenses to choose base or nickel before they knew where he'd line up; the hybrid edge became a position of its own; the read-option wave of 2012 came and went; and analysts showed coaches were kicking on fourth down far too often.
- **Two-high and rotation** (Disguise, Rotation, and Split-Field Coverage): a two-high shell has two safeties deep before the snap; a defense can stay there or rotate one safety down at the snap, so the shell you see isn't always the coverage you get.
- **The box** (A First Look at Defense): the defenders near the line who can stop a run. A light box has no more of them than the offense has blockers.
- **Personnel** (Personnel Groupings): two digits, backs then tight ends, so 11 personnel is one back, one tight end and three receivers, and 12 personnel swaps a receiver for a second tight end.
- **Play-action** (Play-Action, Boots, and Screens): a pass that starts with a fake handoff, so defenders who read run step the wrong way.
- **EPA** (Downs, Distance, and Field Position): expected points added, how much a play changed the offense's expected score. Positive is good for the offense.

Reading the diagrams: offense is blue circles (X, Y, Z, H receivers; Y and U tight ends; F a fullback; Q quarterback; R running back); defense is outlined squares (E and T ends and tackles; M the Mike and W the Will linebacker; \$ the nickel back; CB corners; FS and SS safeties). Solid arrows are routes and runs, dashed arrows defenders' movements, tees blocks. Gold rings mark tight ends and fullbacks, orange rings the defenders to watch, green rings the receiver to watch. A dashed rectangle marks the box. The dark blue line is the line of scrimmage and the orange line the line to gain.

The shape of the era

Start with the whole board. The figure below lays the nine seasons out the way Dynasties and Counterpunches laid out the seventeen before them: offensive moves on the left, defensive answers on the right, rules and new ideas in the middle. Read down a column to follow one side, then across: the answers land a season or two after the moves they answer.

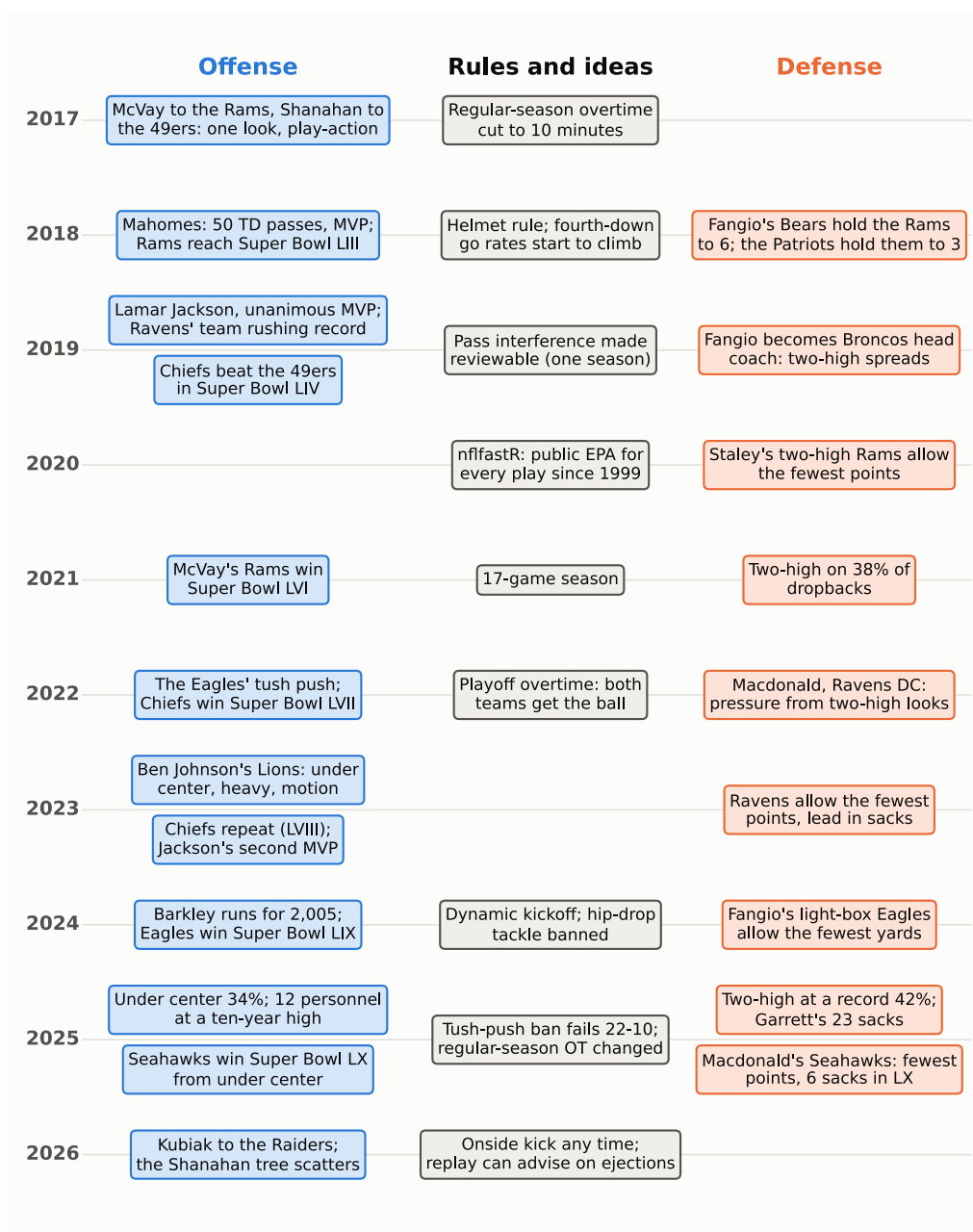


Figure 1: The era at a glance, 2017-2026. Offensive moves (blue, left), defensive answers (orange, right), and the rules and ideas in between (grey, centre). Each item is placed at its season; a Super Bowl is listed under the season it ended (Super Bowl LX, played in February 2026, belongs to 2025). Notice the rhythm: the offenses of 2017-2019 come first, the two-high defenses of 2019-2022 answer them, and the heavy, under-center offenses of 2023-2025 answer those.

Three forces push on every row.

- **The passing game kept its head start.** The rules of the previous two decades had tilted the field toward throwing, and every offense in this chapter is a way of throwing better: off a run fake, after a motion, out of a broken play.
- **Defenses learned to protect the sky.** The single most important defensive change of the era was not a new blitz but a decision about where to put the two safeties, and its cost (fewer defenders near the line) shaped everything offenses did next.
- **Information got cheap.** Tracking data, public play-by-play models and full-time analysts changed how coaches made decisions, most visibly on fourth down.

Now the details, in roughly the order they happened.

2017: two young coaches and an old idea

Picture the league McVay walked into. In 2016 the most imitated defense in football was still Pete Carroll's Seattle Cover 3: a single deep safety in the middle, press-and-bail corners, and a strong safety rolled down near the line as an eighth run defender (the anatomy is in The Cover 3 Family). Carroll's former assistants ran it in Jacksonville and Atlanta, and the Falcons reached Super Bowl LI with it. Offenses, for their part, had mostly moved to the shotgun (about 64 percent of plays in 2016) and spread the field with three receivers.

The Los Angeles Rams were the league's worst offense that year. They scored 224 points, fewest in the NFL. In January 2017 they hired Sean McVay, Washington's 30-year-old offensive coordinator, as head coach, the youngest in the modern NFL. A few weeks later San Francisco hired Kyle Shanahan, the Atlanta coordinator whose offense had just led the league in scoring. The two had coached together in Washington from 2010 to 2013 under Shanahan's father, Mike, and both spoke the same football language: the outside-zone run game and the bootleg passing game that Mike Shanahan and Gary Kubiak had built in Denver in the 1990s (Play-Action, Boots, and Screens traces the tree). In McVay's first season the Rams scored 478 points, the most in the league.²

What did McVay do? Five things, none of them new on its own.

1. **One personnel group.** By 2018 the Rams used 11 personnel on about 93 percent of their snaps, against a league median of about 65. A defense learns a lot from substitutions; when there aren't any, it learns nothing (Constraint Theory and Sequencing calls this the illusion of complexity).
2. **The quarterback under center.** In 2018 Jared Goff took the snap under center on about 61 percent of plays, the most in the league, while the league as a whole did so on about 36 percent.³ Why? With his back to the defense and the back running at full speed past him, a run and a run fake look identical for a beat longer than they do from the shotgun.
3. **The outside zone as the base play.** The whole line steps sideways, and linebackers have to run with it or get cut off (Zone Running).
4. **Play-action and boots built to look exactly like it.** Goff threw off a run fake on about a third of his dropbacks, more than any quarterback in the league.
5. **Tight splits and motion.** The receivers lined up close to the formation (the condensed formation of Spread and Modern Formations), close enough to block a linebacker on a run and to release into routes from the same spot on a pass, and one of them was often in motion before the snap (Motion and Shifts as Weapons).

Here is why those five things together beat the defense of the day. A single-high defense gets its eighth run defender by bringing a safety down, and it asks its linebackers and that safety to attack the run the moment they see it. Against an offense whose runs and passes looked different, that was a fine bet. Against an offense whose runs and passes looked the same for the first second, it meant three defenders stepping forward on every play-action pass, and their steps opened a space on the field that the McVay offense was built to throw into: the **intermediate middle**, 10 to 18 yards downfield, behind the linebackers and in front of the deep defenders.

²Points: nflverse schedules data (Rams 224 points in 2016, fewest in the league; 478 in 2017, most in the league). McVay hired January 12, 2017, at 30, the youngest head coach in modern NFL history: "Sean McVay," Wikipedia: https://en.wikipedia.org/wiki/Sean_McVay ; Shanahan hired by San Francisco in February 2017 after Atlanta's 2016 offense led the league in scoring (540 points): "Kyle Shanahan," Wikipedia: https://en.wikipedia.org/wiki/Kyle_Shanahan . Both on Mike Shanahan's Washington staff 2010-2013. Shotgun share of 2016 runs and passes (64.2 percent): course fact sheet, nflverse play-by-play.

³Under-center share: this book's calculation from nflverse play-by-play, 2018 regular season, runs and passes not flagged as shotgun (Rams 60.9 percent, first in the league; league 36.1 percent). 11 personnel share (about 93 percent, against a league median of about 65): Constraint Theory and Sequencing and Personnel Groupings, from nflverse participation data. Goff's play-action rate (about 35 percent of dropbacks, the league's highest in 2018): Play-Action, Boots, and Screens and its sources.

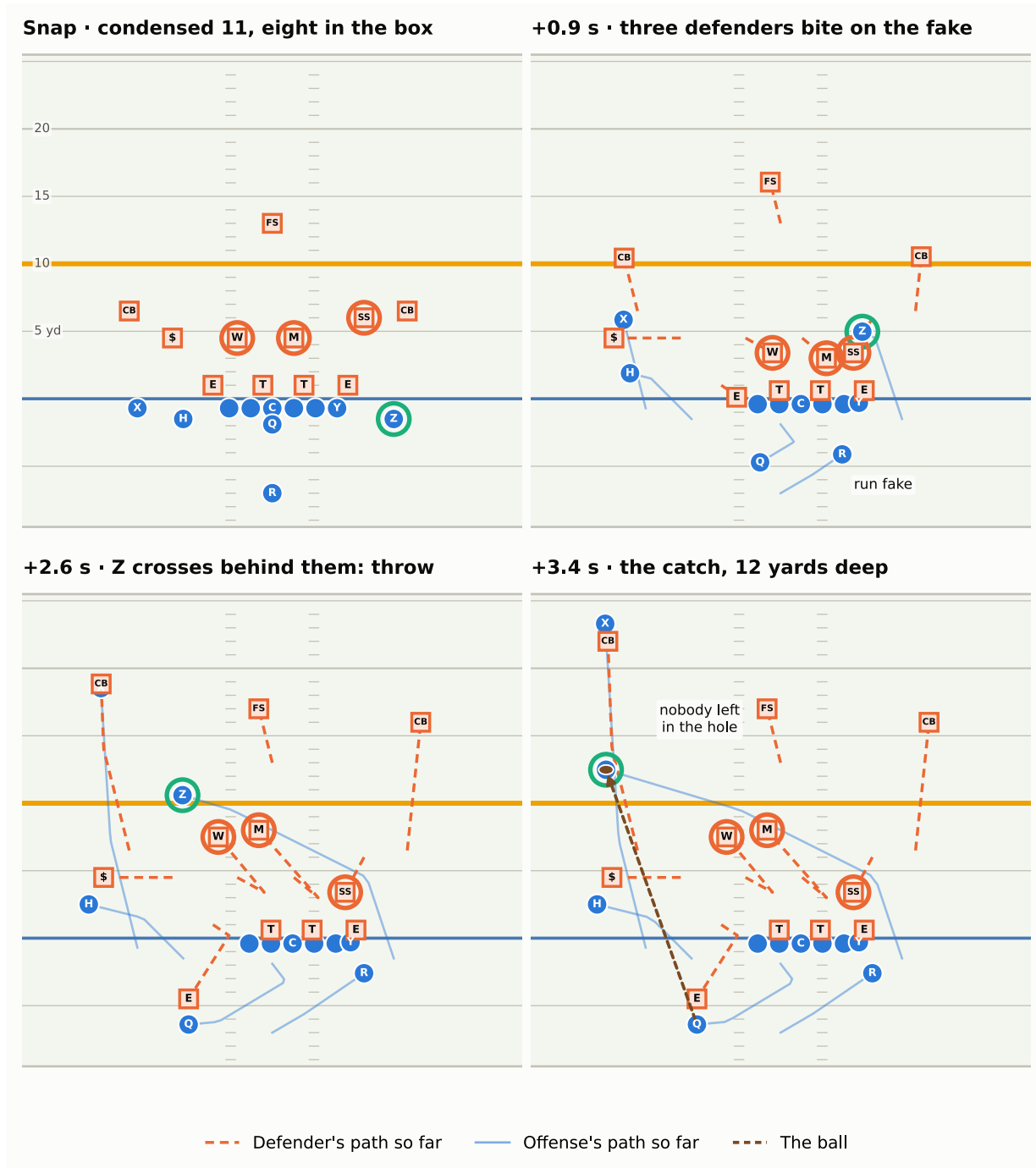


Figure 2: Play-action against single-high, 2018 style (simulated; an illustration of the idea, not a specific Rams play). Snap: 11 personnel in a condensed set, the quarterback under center; the defense is in Cover 3 with eight in the box, the strong safety (SS) rolled down. +0.9 s: the line and the back show outside zone to the right, and the three run-first defenders (ringed: both linebackers and the strong safety) step toward it, as they must against a real run. The quarterback has turned his back to the defense to fake the handoff and is now booting to his left, away from the fake. +2.6 s: Z, from the right side, has crossed behind the linebackers; on the left, X's vertical route has taken the corner deep and H's flat route has pulled the nickel back (\$) wide, and the quarterback, clear of the unblocked end, throws. Last panel: Z catches the deep over about 12 yards downfield near the left numbers, in the hole between the flat and the deep third, with no defender within seven yards.

Follow the three ringed defenders. None of them does anything wrong. The Mike and Will see the line step and the back run toward the sideline, and they come up to fill their gaps; the strong safety, the eighth man the defense brought down for exactly this, attacks the edge. In a Cover 3 defense those three are also the players who guard the middle of the field against passes. By the time they see the quarterback pull the ball back and run the other way, they are four or five yards out of position, and the receiver crossing behind them is running away from all of them. The corner on that side is deep with the vertical route, the nickel back

is wide with the flat route, and nobody is left in the middle. The same logic killed the Tampa 2 Mike in the last chapter: find the defenders with two jobs and make them do the wrong one first.

! Why it exists: the McVay-Shanahan offense

What it solves. A defense reads the offense in a fixed order: personnel, formation, then the first step of the play. McVay's offense made all three tell the defense the same thing every time (11 personnel, a tight formation, an outside-zone step), so the defense had to defend the run honestly and was a beat late on everything else. Motion added a last-second problem, because a defense has to adjust to it and every adjustment leaks information.

What it costs. It needs a run game good enough to fear; play-action against a defense that doesn't respect the run is just a slow dropback with the quarterback's back turned. It asks receivers to block linebackers and safeties. The long fakes and boots take time, so the quarterback is exposed to the unblocked backside end, as in the figure. And the sameness is a two-way street: a defense that can make the run fake *not* matter (by keeping its middle defenders deep no matter what) takes the best part of the offense away. That is exactly what happened next.

i Film room: Rams 2018 — the high point, and the first answers

The 2018 Rams went 13-3, scored 527 points, and reached Super Bowl LIII. On November 19 they beat Kansas City 54-51 in a Monday night game that seemed like a preview of the league's future (Early Downs and Field Zones has it). Then the league's two most respected defensive minds got a look at them. On December 9 in Chicago, Vic Fangio's Bears held the Rams to 6 points and intercepted Goff four times. Eight weeks later, in the Super Bowl, Bill Belichick's Patriots held them to 3 and won 13-3.⁴

What to notice: in both games, watch the second level on the Rams' run-action passes. The Bears and the Patriots didn't let the fake pull their linebackers and safeties forward the way the figure above does; defenders sank toward the intermediate middle, where the crossers were going, and the pass rush got to Goff before the long-developing routes came open. Fangio's way of doing that, safeties who stay deep and a front that holds the run without them, is the seed of the rest of this chapter. Belichick's plan has its own diagram in Belichick and Game-Plan Football.

Shanahan's 49ers took a slightly different road to the same place. Where McVay lived in 11 personnel, Shanahan used a fullback, Kyle Juszczyk, and a tight end, George Kittle, who could each block and catch, so he could show heavy personnel and still throw. His 2019 team went 13-3 and reached Super Bowl LIV. The case study The Shanahan Tree follows that offense and the coaches it produced, who by 2025 were calling plays all over the league.

Mahomes, Reid, and the play after the play

The other offensive revolution of 2018 came from Kansas City, and it worked on a different axis. Andy Reid had been a head coach since 1999, a West Coast offense coach from Mike Holmgren's Green Bay staffs who had spent two decades building passing games out of short, timed throws and screens. In 2017 the Chiefs traded up to draft Patrick Mahomes tenth overall out of Texas Tech, where he had played in Kliff Kingsbury's Air Raid, the pass-everything college system of four- and five-receiver sets and a few concepts practised endlessly. Mahomes sat for a season, then in 2018, his first as a starter, threw for 5,097 yards and 50 touchdowns and was named MVP. The Chiefs scored 565 points, the most in the league.⁵

Two things made the Reid-Mahomes offense different from McVay's.

It borrowed from college, openly. Reid's staffs were known around the league for studying college film and lifting whole ideas from it: the RPO (a run with a pass option attached, decided by reading one defender; RPOs), the shovel option, jet motion, and the Air Raid concepts Mahomes already knew. Where McVay made every play look like one run, Reid made his offense look like many different college offenses, often in the same game.

⁴nlverse schedules and play-by-play: December 9, 2018, Rams at Bears, Chicago 15, Los Angeles 6, with four Goff interceptions (Roquan Smith, Eddie Jackson, Kyle Fuller, Prince Amukamara); Super Bowl LIII, February 3, 2019, Patriots 13, Rams 3. November 19, 2018: Rams 54, Chiefs 51. Rams 2018: 13-3, 527 points. Fangio was Chicago's defensive coordinator 2015-2018: "Vic Fangio," Wikipedia: https://en.wikipedia.org/wiki/Vic_Fangio.

⁵"Patrick Mahomes," Wikipedia: https://en.wikipedia.org/wiki/Patrick_Mahomes (drafted 10th overall in 2017 after a trade up from 27th; Texas Tech 2014-2016 under Kliff Kingsbury; 2018: 5,097 passing yards, 50 touchdown passes, AP MVP). Chiefs' 565 points in 2018, the most in the league: nlverse schedules data. Reid's background (Holmgren's Packers staffs; Eagles head coach from 1999; Chiefs from 2013): "Andy Reid," Wikipedia: https://en.wikipedia.org/wiki/Andy_Reid.

Its best plays were often not the plays that were called. Mahomes's gift was the off-script play (Quarterback Play): the play after the design breaks down, when the quarterback escapes the rush, keeps his eyes downfield and throws from an angle no defense practises against, including, once, left-handed. A defense can study a play-caller's tendencies. It cannot study a scramble.

That combination is why Kansas City matters to this chapter's arms race. McVay's offense punished a defense that put too many players near the line. Mahomes punished a defense that left any receiver alone deep, because given an extra second and a broken pocket he would find him. Against both at once, the natural defensive answer was the same: keep more players back, deep, with everything in front of them. By 2021, by this book's count from nflverse charting data, defenses played a two-high coverage against the Chiefs on about 46 percent of labelled dropbacks, the third-highest share against any offense and well above the league's 38 percent.⁶ The Chiefs' answer to that answer was patience: fewer deep shots, more of the short game Reid had always loved, Travis Kelce underneath, and Mahomes's legs when the rush came. It worked. From the 2019 season to the 2024 season Kansas City played in five of six Super Bowls and won three of them.⁷

i Film room: Super Bowl LIV, February 2, 2020 — third-and-15

The Chiefs trailed San Francisco 20-10 with 7:13 left in the fourth quarter and faced third-and-15 at their own 35. Mahomes took a deep drop, held the ball while Tyreek Hill ran a long, deep-breaking route that the Chiefs called "Jet Chip Wasp," and threw it 44 yards to Hill at the San Francisco 21. Kansas City scored three touchdowns in the last seven minutes and won 31-20.⁸

What to notice: both halves of the offense in one snap. The design is Reid's: a route built to attack a defense that was playing deep and soft on third-and-long, and a protection (the "chip": a back bumping the edge rusher on his way out) built to buy time. The time is Mahomes's: he holds the ball past the moment most quarterbacks would check it down. Defending that offense meant defending both, and the cheapest way to do it was to put more players deep.

The two-high counterrevolution

By 2019 defensive coaches had two problems: play-action that pulled their middle defenders forward, and quarterbacks who punished any single-covered receiver deep. Both problems had the same root. A single-high defense commits its eighth defender to the run *before the snap*, and modern offenses had learned to make that commitment cost something on every play.

The answer that swept the league is what this book calls the **two-high counterrevolution**: defenses that line up with two safeties deep, play a two-deep coverage (quarters, Cover 6, Cover 2) or rotate out of that shell only at the snap, and accept fewer defenders near the line in exchange. It is a "counter" revolution because it reverses the previous decade's direction. Seattle's era had pulled safeties down toward the line; this pushed them back.

Its roots are in college. As Disguise, Rotation, and Split-Field Coverage tells it, quarters defenses at Michigan State, TCU and Nick Saban's schools spent the 2000s learning to play the run with safeties at 10 to 12 yards. In the NFL the coach most associated with it is **Vic Fangio**, a defensive assistant for more than thirty years, the Bears' coordinator from 2015 to 2018 and, at 60, a first-time head coach in Denver from 2019. Fangio's defenses kept two safeties deep, played quarters-based coverages that read the receivers in front of them, and disguised which coverage was coming until the snap. His protégé **Brandon Staley** took the system to the Rams in 2020, and with Aaron Donald inside and Jalen Ramsey moving around the secondary, the Rams allowed the fewest points in the league. Staley was a head coach a year later, and coaches from Fangio's and Staley's staffs (Joe Barry, Sean Desai, Ejiro Evero, Chris Shula) spread the approach from there.⁹ Fangio himself coordinated the Eagles' defense to a Super Bowl title in the 2024 season.

⁶This book's calculation from nflverse participation data joined to play-by-play: the share of each offense's dropbacks with a charted coverage on which the defense played Cover 2, 2-Man, Cover 4 or Cover 6. Kansas City faced two-high coverage on 45.9 percent in 2021 (third, behind Pittsburgh at 48.5 and the Rams at 46.7; league 38.4 percent). Coverage labels exist only from 2018 and change source between 2022 and 2023 (course fact sheet §7).

⁷nflverse schedules data and "Kansas City Chiefs," Wikipedia: https://en.wikipedia.org/wiki/Kansas_City_Chiefs. Super Bowls LIV (beat San Francisco 31-20), LV (lost to Tampa Bay 31-9), LVII (beat Philadelphia 38-35), LVIII (beat San Francisco 25-22 in overtime) and LIX (lost to Philadelphia 40-22), after the 2019, 2020, 2022, 2023 and 2024 seasons.

⁸nflverse play-by-play of Super Bowl LIV (February 2, 2020): fourth quarter, 7:13, third-and-15 at the Kansas City 35, "P.Mahomes pass deep left to T.Hill to SF 21 for 44 yards"; San Francisco led 20-10 at the time; final 31-20. "Super Bowl LIV," Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_LIV.

⁹"Vic Fangio," Wikipedia: https://en.wikipedia.org/wiki/Vic_Fangio (NFL assistant from 1986; Bears defensive coordinator 2015-2018; Broncos head coach 2019-2021, hired at 60; Dolphins defensive coordinator 2023; Eagles defensive coordinator from 2024, still in 2026 per the course fact sheet). The 2020 Rams allowed 296 points, fewest in the league (nflverse schedules data); Staley was the Chargers' head coach 2021-2023: "Brandon Staley," Wikipedia: https://en.wikipedia.org/wiki/Brandon_Staley. The tree: Diante Lee, "Checking in on the Vic Fangio coaching tree after Week 1 of the 2021 NFL season," PFF (September 15, 2021): <https://www.pff.com/news/nfl-vic-fangio-coaching-tree-week-1-2021>; Evero and Shula: Disguise, Rotation, and Split-Field Coverage and the course fact sheet (2026 staffs).

The idea has three parts.

1. **Keep a lid on the field.** Two deep safeties take away the explosive pass, and because they sit at 10 to 14 yards reading the receivers in front of them, they also sit right where McVay's crossers and Mahomes's deep shots wanted to go. The run fake loses its prize: a safety reading a receiver doesn't care what the back is doing.
2. **Let the front hold the run, and fit it late with the safeties.** With both safeties back, the box usually has six defenders against six blockers (five linemen and a tight end). The defensive line has to win its own fights, and the safeties come downhill to make the tackle after the runner is past the line, at five or six yards instead of two.
3. **Don't tell the quarterback.** A two-high shell is the best starting point for disguise, because either safety can rotate down at the snap and turn it into a single-high coverage, or neither does (post-snap rotation in 06-06). The quarterback who reads "two-high" before the snap may be wrong a second later.

The second part is the bargain, and it deserves a picture.

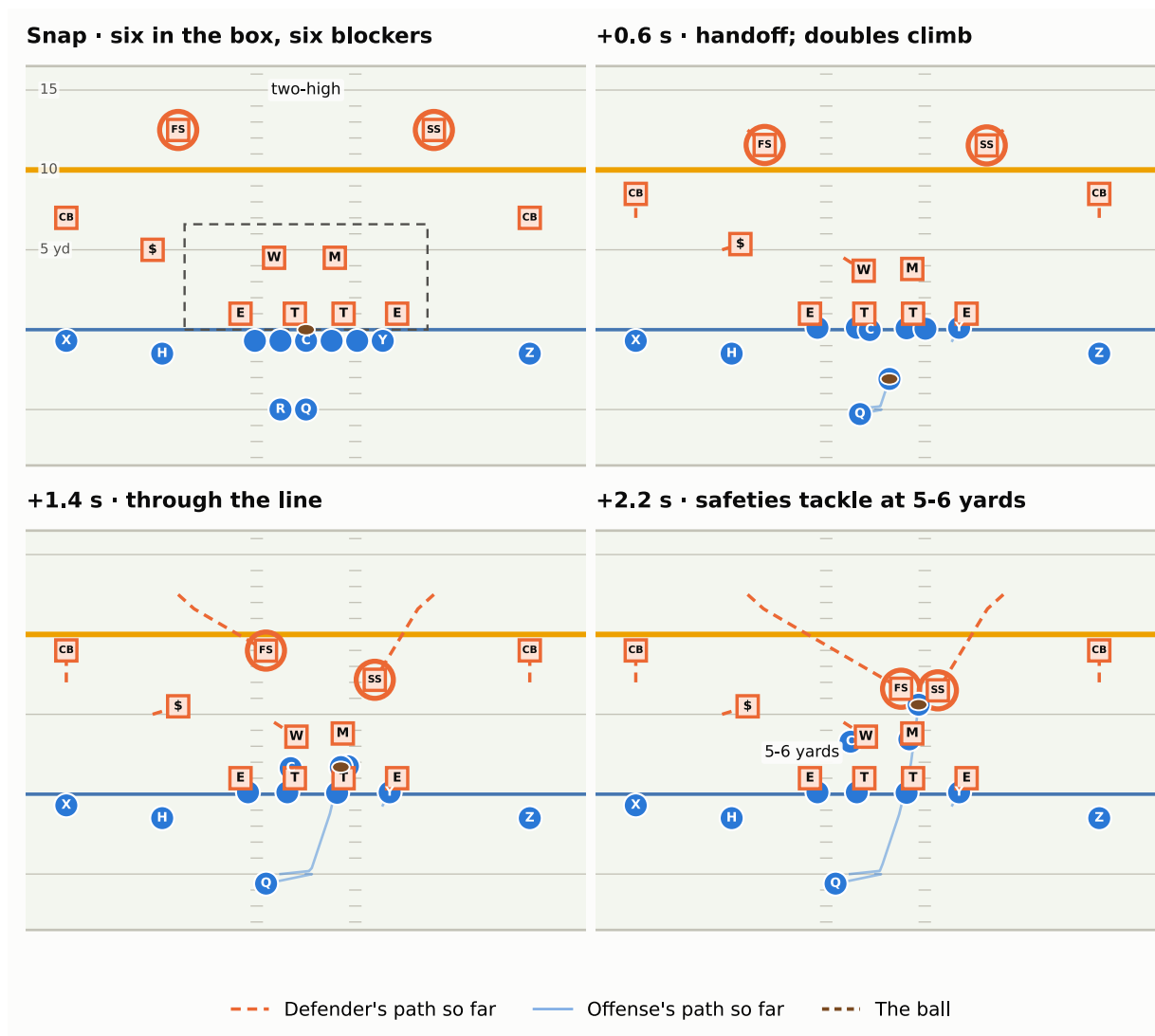


Figure 3: The bargain, simulated: an inside-zone run against a two-high defense with six in the box (an illustration, not a specific play). Snap: 11 personnel in the shotgun with the tight end attached on the right; the defense has four linemen and two linebackers in the box (dashed rectangle) and both safeties (ringed) at about 12 yards. The offense has six blockers, five linemen and the tight end, so every box defender has a man. +0.6 s: the handoff; the linemen double-team the two defensive tackles and climb to the linebackers. +1.4 s: the back is through the line untouched. +2.2 s: the two safeties arrive from depth and make the tackle about 5 or 6 yards downfield. That is a good run for the offense, and it is exactly the run the defense agreed to give up.

Five yards on a run looks like a win for the offense, and it is a small one. But think about what the defense bought with it. In 2025, early-down runs against six or fewer box defenders averaged about 4.6 yards; against seven or more, about 4.1. In expected points, runs against light boxes were still slightly *negative* on average (about -0.04 EPA per play), just less negative than runs

against heavier boxes (about -0.09).¹⁰ Meanwhile the thing the defense was protecting against, the 20-yard pass and the 50-yard touchdown, got rarer: quarterbacks threw 20 or more yards downfield on about 10 percent of their attempts against two-high coverages in 2025, against about 11 percent against single-high ones. A defense that gives up an extra half-yard per run to take away a few deep balls a game is making a trade the numbers like, as long as its front can hold.

Here is the whole era in the data.

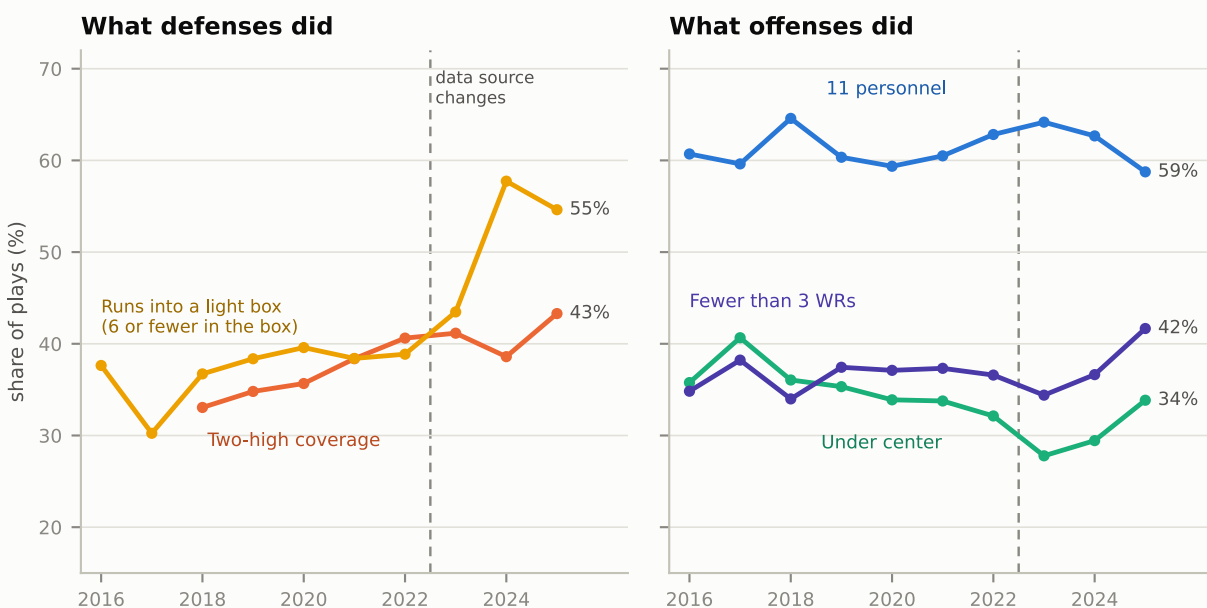


Figure 4: The arms race in five lines, 2016-2025 (regular seasons; this book's calculation from nflverse play-by-play and participation data). Left, what defenses did: two-high coverages (Cover 2, 2-Man, quarters and Cover 6) as a share of dropbacks with a charted coverage, which starts in 2018 because 2016-2017 have no labels; and the share of designed runs that faced six or fewer defenders in the box. Right, what offenses did: 11 personnel, snaps not in the shotgun (almost all under center; a few pistol), and snaps with fewer than three wide receivers. The dashed line marks a change of data source (Next Gen Stats through 2022, FTN charting from 2023), so read jumps across it with care; Next Gen Stats' own two-high series, which doesn't cross a source change, climbs from 32.9% in 2018 to 42.0% in 2025. Two-high climbs steadily from 2018; light boxes follow; then, in 2025, offenses get heavier and go back under center.

Read it left to right, the way the coaches lived it.¹¹

- **Two-high climbs.** In the nflverse charting, two-high coverages went from about 33 percent of dropbacks in 2018 to about 43 percent in 2025. Next Gen Stats, measuring the same thing with its own tracking, has 32.9 percent in 2018, 37.8 in 2021, 40.4 in 2024 and a record 42.0 percent in 2025, when every defense in the league was above 30 percent and quarters alone was 17 percent of coverages.¹²
- **The boxes empty.** Runs against six or fewer defenders went from about 37 percent of designed runs in 2018 to more than half in 2024 and 2025. (Part of the jump in 2024 may be how the new charting service counts the box; the direction matches every other source.)

¹⁰This book's calculation from nflverse play-by-play and participation data, 2025 regular season: first- and second-down designed runs (no scrambles) by the number of defenders in the box. Six or fewer: 4.64 yards per carry, -0.042 EPA per play; seven or more: 4.09 yards, -0.085 EPA. Every season 2016-2025 shows the same pattern (light-box runs about half a yard better). Deep attempts (20 or more air yards) as a share of pass attempts with a charted coverage, 2025: 10.0 percent against two-high coverages, 11.3 percent against single-high (Cover 0, 1, 3 and 9); in 2024, 8.4 and 12.0 percent.

¹¹This book's calculation, reproducible from the code above the figure: nflverse play-by-play (shotgun, qb_dropback) joined to load_participation() (offense_personnel, defenders_in_box, defense_coverage_type), regular-season runs and passes with a recorded down. Two-high: 33.1 (2018), 38.4 (2021), 40.6 (2022), 41.2 (2023), 38.6 (2024), 43.3 percent (2025). Runs facing six or fewer in the box: 36.7 (2018), 43.5 (2023), 57.7 (2024), 54.6 percent (2025). 11 personnel: 64.6 (2018), 64.2 (2023), 58.8 percent (2025). Not in the shotgun: 36.1 (2018), 27.8 (2023), 33.8 percent (2025); FTN's own under-center flag gives 27.3 (2023), 28.8 (2024) and 33.6 percent (2025). Fewer than three wide receivers: 34.0 (2018), 34.4 (2023), 41.7 percent (2025). Participation data come from Next Gen Stats through 2022 and FTN from 2023 (course fact sheet §7).

¹²NFL.com / Next Gen Stats, "The coverage that turned the Seahawks into Super Bowl winners and is taking the NFL by storm" (2026): <https://www.nfl.com/news/the-coverage-that-turned-the-seahawks-into-super-bowl-winners-and-is-taking-the-nfl-by-storm> (split-safety rate 32.9 percent in 2018, 37.8 in 2021, 40.4 in 2024 and 42.0 in 2025, the highest since tracking began; every defense above 30 percent in 2025; Cover 4 at 17.2 percent of coverages in 2025, against a 13.9 percent average in 2018-2024).

- **Offenses keep three receivers out, until they don't.** 11 personnel barely moved from 2016 to 2024, around 60 to 64 percent of plays. Then, in 2025, it fell to its lowest share of the decade, and snaps with fewer than three receivers jumped to about 42 percent. Shotgun use peaked in 2023 at about 72 percent of plays, and under-center snaps came back, from about 28 percent in 2023 to about 34 in

2025.

The left panel is the counterrevolution. The right panel is the counter to the counter, and its last season is the most interesting season in the chart.

Common misconception

“Two-high is a soft, prevent-style defense.” It looks soft before the snap, with safeties 12 yards deep and corners off. But the two-high defenses of this era were built to *move*: a quarters safety is the first run defender to the edge on his side, a rotating safety can arrive as a robber in the middle of the field at the snap, and the shell is a disguise for single-high coverages and pressures as often as it is a coverage in itself. The 2020 Rams, 2023 Ravens and 2024 Eagles, three of the era’s two-high defenses, each allowed the fewest points or yards in the league. Soft defenses don’t do that.

The offense answers: bigger, tighter, and under center

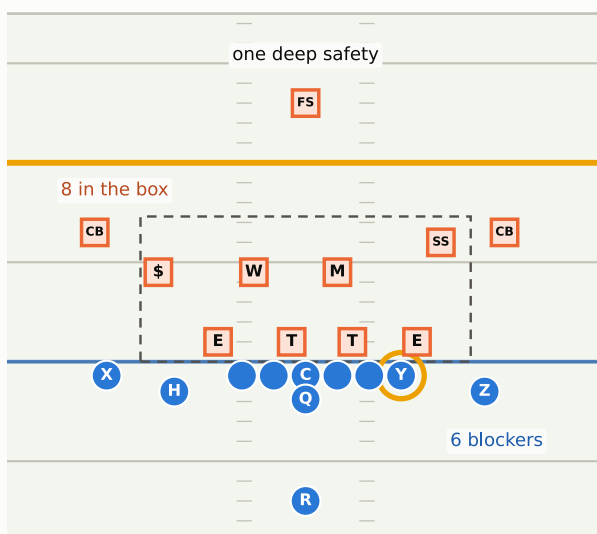
If the defense keeps two safeties deep and six in the box, the offense’s first question is simple arithmetic. How many blockers can it put against those six?

From 11 personnel the answer is six: five linemen and a tight end. Six against six means every defender is blocked if everyone wins, but nobody is left over to handle a missed block, and the runner has to make the last defender miss. Now take a receiver off the field and put on a second tight end, or a fullback. Seven blockers against six. The defense has three choices, and none of them is free:

- **Stay in its nickel, two-high shell** and be outnumbered in the box. The offense runs.
- **Bring a safety down**, at the snap or before it, to get to seven. Now it is single-high again, and the offense is back in McVay’s world, with play-action against defenders who must respect the run.
- **Substitute bigger players**, a third linebacker or a third safety, to match the heavy personnel. Now the offense can spread those bigger players out and throw at the linebacker covering a tight end, the move tight end’s dilemma from Dynasties and Counterpunches.

That arithmetic is the **light-box run game**: an offense built to run into the boxes two-high defenses leave, with enough blockers that the defense must either concede the run or give up its shell to stop it. The opening of this chapter shows the two pictures side by side.

2018: 11 personnel vs. one-high



2025: 12 personnel vs. two-high

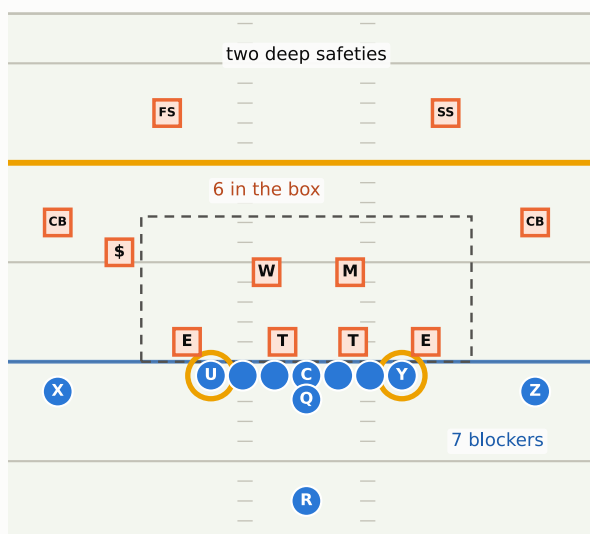


Figure 5: Nine seasons apart (illustrative alignments). Left, a 2018-style snap: 11 personnel in a condensed set, the quarterback under center, against a single-high defense with the strong safety (SS) rolled down, eight defenders in the box (dashed rectangle) and one safety deep. Right, a 2025-style snap: 12 personnel (two tight ends, gold rings) under center, against a nickel two-high defense with six in the box and two safeties deep. In 2018 the defense outnumbers the offense's blockers near the line (eight defenders, six blockers) and the offense throws behind them; in 2025 the offense outnumbers the defense (seven blockers, six defenders) and dares it to bring a safety down.

Offenses took the second and third answers seriously, and the 2025 numbers in the trend chart are the result. 12 personnel reached 24 percent of plays, its highest share in the decade of data, and 13 personnel (three tight ends) 5 percent; Next Gen Stats counted fewer than three wide receivers on 41.7 percent of plays, the first season at or above 38 percent since it began tracking in 2016. Defenses answered in kind: they played base defense on 29.7 percent of snaps, up from a low of 20.5 percent in 2023.¹³ Some offenses went further than others. The 2025 Rams, McVay's team, the same coach who had lived in 11 personnel in 2018, used three tight ends on about 31 percent of their snaps. Shanahan's 49ers had a fullback on the field for 37 percent of theirs.¹⁴

Why under center?

The trend chart's green line is the other half of the answer, and it is the one most fans notice first, because it changes what the quarterback looks like before the snap. This book calls it the **under-center revival**: the quarterback taking the snap directly behind the center on about a third of plays in 2025, up from about a quarter in 2023, with Shanahan-tree offenses leading. The 2025 Rams were under center on 59 percent of their plays, the Seahawks on 53, and the Bills, Lions, Bears and 49ers between 46 and 50 percent, against a league average of 34.

Why would an offense give up the shotgun, the alignment that lets the quarterback see the whole defense and throw quickly? Four reasons, each tied to the two-high defense.

1. **The run gets better.** On an outside-zone or duo run from under center, the back takes the ball already running downhill at full speed, aiming wherever the blocks open; from the shotgun he starts beside the quarterback and has to run sideways first. Against a light box, the offense wants the run to be as good as it can be.
2. **The fake gets better.** From under center the quarterback turns his back to the defense to hand off or to fake it, and the linebackers and safeties can't see the ball. From the shotgun the fake is in plain view.
3. **The safety has to decide.** This is the big one. In a two-high defense the safeties are the extra run defenders: one of them has to come down and fit the run, late, from depth. A run that looks the same as play-action for a full second makes that safety guess. If he comes down, the play-action pass goes behind him. If he waits, the run gets its five yards.

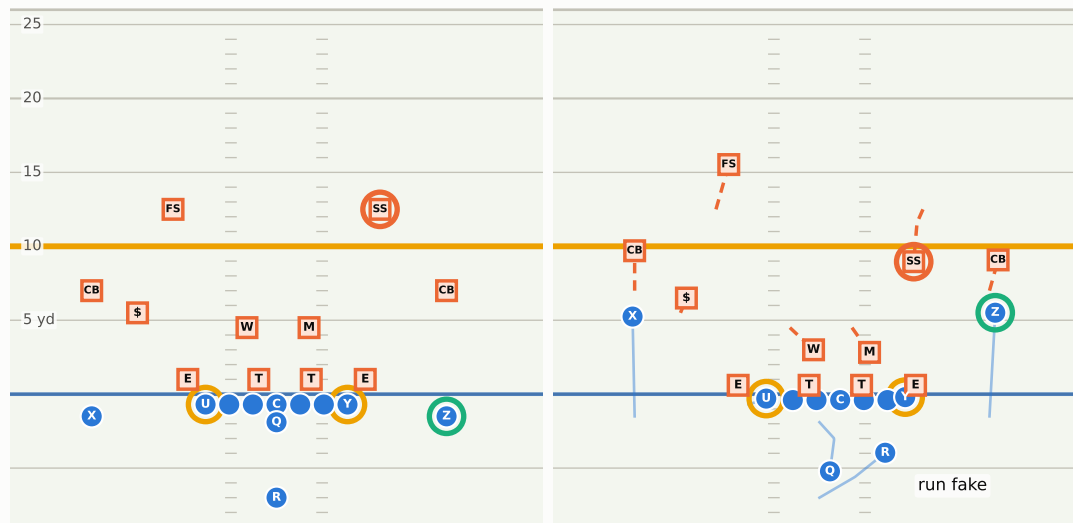
¹³James Reber, NFL.com, "NFL trend watch: Heavy personnel could shape Super Bowl LX showdown between Patriots and Seahawks" (2026), from Next Gen Stats: <https://www.nfl.com/news/nfl-trend-watch-heavy-personnel-could-shape-super-bowl-lx-showdown-between-patriots-and-seahawks> (fewer than three wide receivers on 41.7 percent of plays in 2025, the first season at or above 38 percent since 2016; base defense on 29.7 percent of snaps in 2025, against 20.5 percent in 2023). 12 personnel (24.4 percent) and 13 personnel (5.1 percent) in 2025: course fact sheet, nflverse participation.

¹⁴This book's calculation from nflverse participation and FTN charting data, 2025 regular season, runs and passes: Rams 13 personnel 31.3 percent of plays, under center 59.1 percent (FTN, first in the league); 49ers 21 personnel 37.1 percent. Other under-center shares: Seattle 53.4, Buffalo 49.5, Detroit 48.0, Chicago 47.5, San Francisco 45.8; league 33.6.

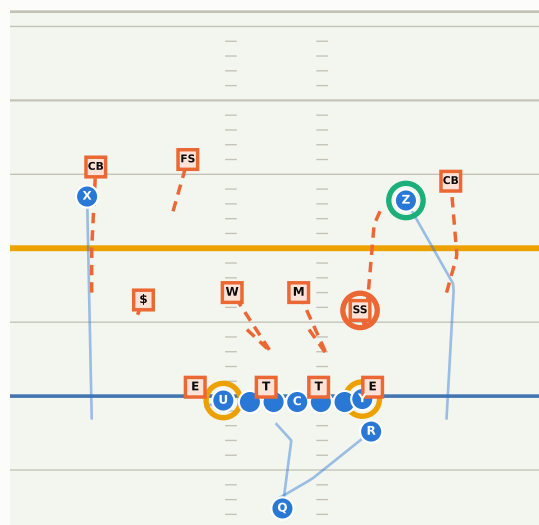
4. **The pre-snap picture tells the defense less.** In the shotgun, a back offset to one side tells a defense something about which way a run can go and who is staying in to block. A back seven yards behind a quarterback under center can go anywhere.

The cost is real. A quarterback under center takes longer to set up and sees the defense later; a dropback from under center is slower than one from the gun; and on obvious passing downs the shotgun still wins, which is why third-and-long remains a shotgun down for almost every team. The revival is a first-and-second-down weapon.

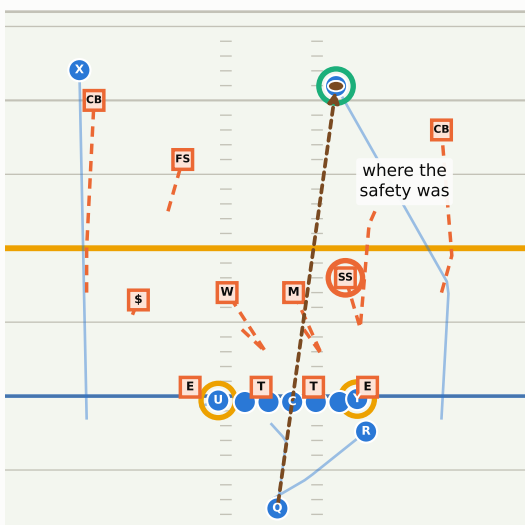
Snap · 12 personnel under center, two-high · 0.9 s · the strong safety fits the 'run'



+2.0 s · Z breaks behind him: throw



+3.1 s · the catch, 22 yards deep



--- Defender's path so far — Offense's path so far - - - The ball

Figure 6: The payoff, simulated: play-action from 12 personnel under center against a two-high (quarters) defense with six in the box (an illustration of the idea, not a specific team's play). Snap: two tight ends attached (gold rings), the quarterback under center; two safeties at 12 yards. +0.9 s: both tight ends block and the back runs the outside-zone fake to the right. To the strong safety (SS, orange ring), reading the tight end in front of him, it is a run, and in a light box he is the man who must fit it, so he comes down hard. +2.0 s: Z has run past the corner's outside leverage and broken to the post, into the space the strong safety just left; the quarterback, finished with his fake and his drop, throws. Last panel: the catch, about 22 yards downfield, with the safety still turning around. Against a light box the run is good enough that the safety has to come; the under-center fake makes the pass look like that run.

Compare this play with the 2018 one. The geometry is the same: a run fake pulls a defender who guards the middle forward, and the ball goes behind him. What changed is *who* gets pulled. In 2018 it was the linebackers and a box safety, who were always

going to attack the run. In 2025 it is a deep safety, who is only attacking the run because the box is light and the offense has more blockers than the defense has bodies. The offense had to earn that by being good enough at running into light boxes that the safety couldn't wait.

Common misconception

"The under-center comeback means football is going back to the 1990s." The pictures rhyme, but almost everything inside them is new. The 1990s offense lined up under center because that was where quarterbacks stood; the 2025 offense does it to beat a specific defense, and it still runs RPOs, motion at the snap and spread passing concepts from the same alignment. The 1990s defense played an eighth man in the box against two tight ends; the 2025 defense keeps two safeties deep against them and fits the run from 12 yards. Even the fullback is different: the modern one is as likely to run a route as to lead-block. History doesn't repeat here. It gets folded into the next scheme.

Film room: Eagles 2024 — both halves of the argument on one roster

The 2024 Philadelphia Eagles are the clearest single team to watch for this section, because they played both sides of it. On defense, Vic Fangio ran his two-high, light-box system and the Eagles allowed the fewest yards in the league (Run Fits and Box Math has the film room on how the front held up). On offense, Saquon Barkley ran for 2,005 yards behind one of the best lines in football, and about 61 percent of his carries came against six or fewer defenders in the box, where he averaged 5.9 yards. Jalen Hurts's tush push (below) handled the short yardage. The Eagles beat Kansas City 40-22 in Super Bowl LIX, with a four-man rush that never needed to blitz Mahomes.¹⁵

What to notice: on an Eagles run, count the box before the snap. Then watch the opposing safeties: if the box is light and the run gets to them, they make the tackle at five or six yards, and the Eagles are happy to keep doing that all afternoon. When a safety starts creeping down to stop it, watch for the play-action shot behind him.

Positions mutate

An arms race changes the soldiers as well as the weapons. Over nine seasons, six positions changed shape, and every change is an answer to something earlier in this chapter. The map below puts them on one field.

¹⁵Saquon Barkley's 2,005 rushing yards in 2024: "Saquon Barkley," Wikipedia: https://en.wikipedia.org/wiki/Saquon_Barkley. Super Bowl LIX (February 9, 2025): Eagles 40, Chiefs 22, Hurts MVP, pressure from four-man rushes (course fact sheet §1). Fewest yards allowed in 2024 and Barkley's carries against light boxes (61.4 percent of 345 carries against six or fewer in the box, 5.87 yards per carry; 5.71 against seven or more): this book's calculation from nflverse play-by-play and participation data.

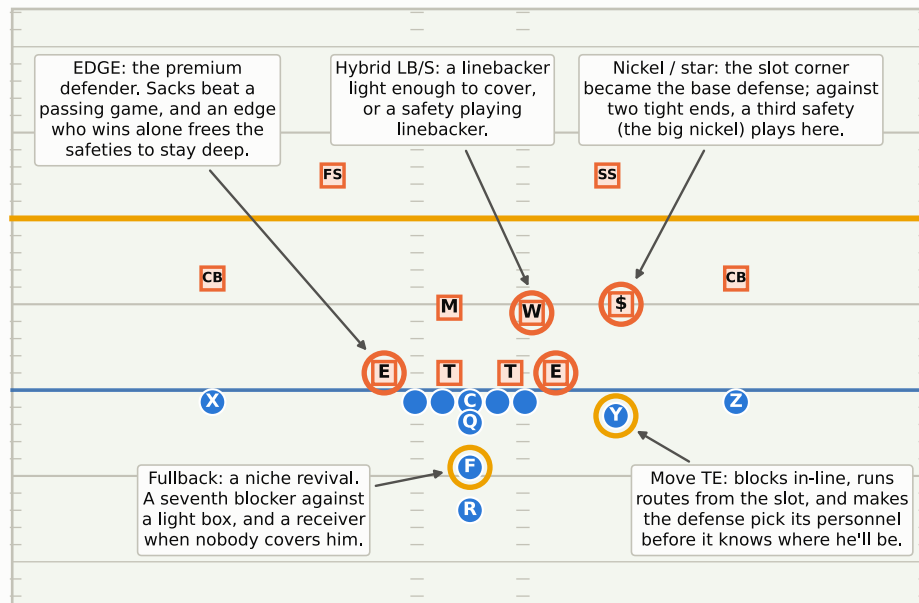


Figure 7: Where the positions changed, 2017-2025 (illustrative alignment). Offense in 21 personnel with a move tight end (Y) split into the slot and a fullback (F) in the backfield (gold rings); defense in nickel with two deep safeties, its edge rushers (E), its hybrid linebacker (W) and its nickel or star (the \$) ringed. Each label says what the position became and which move in the arms race made it that way.

Take them one at a time.

The slot corner, and the nickel as the base defense. When offenses went to three receivers on most snaps, the third receiver needed a third cornerback, the slot cornerback or nickel back. Through this era defenses had five or more defensive backs on the field for roughly two-thirds to three-quarters of snaps (about 66 percent in 2017 and 78 percent in 2023, then back to 69 percent in 2025 as offenses got heavier).¹⁶ The nickel back stopped being a substitute and became a starter, and the “base” 4-3 or 3-4 became the package a team played on a minority of snaps (A First Look at Defense charts it).

The star, and the big nickel. Then the offenses got heavier again, and a 185-pound slot corner stuck in the box against two tight ends became a target. The answer was a safety in the nickel spot: the big nickel, whose fifth defensive back is a third safety big enough to fit a run and fast enough to cover a tight end. Defenses call him the “star,” and the best of them, such as Kyle Hamilton in Mike Macdonald’s Ravens defense or Brian Branch in Detroit, line up deep, in the slot, on the line and in the box, sometimes in the same series. Seattle answered the move tight end this way a decade earlier with Kam Chancellor; the two-high era made it standard.

The hybrid linebacker and safety. Linebackers had to cover more and more space: tight ends, backs in the flat, crossers behind them. Teams drafted lighter, faster linebackers and moved safeties down to play the position. Not every experiment worked: Isaiah Simmons, drafted eighth overall in 2020 as a “positionless” linebacker-safety, never found a home in Arizona’s defense.¹⁷ But the direction held. In a two-high defense with six in the box, the two linebackers have more ground to cover against the run *and* the pass than anyone in the old 4-3, and the players who can do both are the ones who play.

The move tight end. The receiving tight end of the last chapter became the centre of offenses: Travis Kelce in Kansas City, George Kittle in San Francisco, Mark Andrews in Baltimore, and a younger generation such as Brock Bowers, who set the record for catches by a rookie in 2024.¹⁸ He is the reason 12 personnel worked as an answer to two-high: one personnel group that can be a seven-blocker run formation or a spread passing formation, with the defense choosing its players first.

¹⁶This book’s calculation from nflverse participation data ([defense_personnel](#)), regular-season runs and passes: five or more defensive backs on 65.7 percent of plays in 2017, 74.3 in 2018, 78.1 in 2023 and 69.0 in 2025. The full chart by package is in A First Look at Defense.

¹⁷“Isaiah Simmons,” Wikipedia: https://en.wikipedia.org/wiki/Isaiah_Simmons (eighth overall pick, 2020 draft, by Arizona; a Clemson defender who played linebacker, safety and slot corner; traded to the Giants in 2023). Kyle Hamilton (Ravens, first round 2022) and Brian Branch (Lions, second round 2023) as multi-position defensive backs: “Kyle Hamilton (American football),” Wikipedia: [https://en.wikipedia.org/wiki/Kyle_Hamilton_\(American_football\)](https://en.wikipedia.org/wiki/Kyle_Hamilton_(American_football)); “Brian Branch,” Wikipedia: https://en.wikipedia.org/wiki/Brian_Branch.

¹⁸“Brock Bowers,” Wikipedia: https://en.wikipedia.org/wiki/Brock_Bowers (112 receptions as a Raiders rookie in 2024, the NFL record for a rookie).

The edge rusher, the premium defender. The edge rusher, the hybrid edge of the last chapter, became the most valuable defender in the game, and for a reason this chapter can now state precisely. A two-high defense needs pressure without blitzing, because every blitzer comes out of the coverage it is trying to keep. An edge who wins on his own is what lets the safeties stay deep. The market priced it: in 2025, Myles Garrett, T.J. Watt and Micah Parsons all signed contracts worth about \$40 million a year or more, the richest ever for non-quarterbacks, and Garrett then set the single-season sack record with 23.¹⁹

The fullback, a niche revival. The fullback almost disappeared in the spread era. He came back, in small numbers, in the offenses that wanted a seventh blocker against light boxes: Shanahan's 49ers (Kyle Juszczyk, signed in 2017 to what was then by far the richest fullback contract in the league), the Ravens (Patrick Ricard), and the 2025 Seahawks.²⁰ League-wide, two-back personnel stayed around 7 or 8 percent of plays. The revival is in what he does, not how often he plays: the modern fullback motions, catches passes and lines up as a tight end, which is why defenses can't just treat him as a lead blocker and load the box.

The quarterback runs, and this time it sticks

Dynasties and Counterpunches ended the 2012 read-option wave with a line: the wave receded; the tide came in later. This is the tide.

It came in with **Lamar Jackson**. The Ravens drafted him with the last pick of the first round in 2018, and when he took over midseason they rebuilt the offense around him under coordinator Greg Roman: a run game with a quarterback who was a real runner on every snap, heavy personnel, and option reads from the pistol and the shotgun. In 2019 Jackson ran for 1,206 yards, then the record for a quarterback, threw 36 touchdown passes, and was the unanimous MVP; the Ravens ran for 3,296 yards, the most by any team in NFL history.²¹ He won a second MVP in 2023.

Then the others. **Jalen Hurts** led the Eagles to two Super Bowls (the 2022 and 2024 seasons) with designed runs, scrambles and the tush push, the quarterback sneak with teammates shoving from behind that the Eagles turned into a near-automatic first down (Red Zone, Goal Line, and Short Yardage has it). **Josh Allen**, a 6-foot-5 quarterback with a cannon for an arm, ran for double-digit touchdowns in multiple seasons, won the MVP in 2024 and, by the end of 2025, had more career rushing touchdowns than any quarterback in history. Jayden Daniels ran for more than 850 yards as a rookie in 2024 and took Washington to the NFC Championship Game.²²

¹⁹Contracts: Myles Garrett (Browns, March 2025, about \$40 million a year), T.J. Watt (Steelers, July 2025, about \$41 million a year) and Micah Parsons (traded to Green Bay in August 2025 and signed at about \$47 million a year), each at signing the richest average salary for a non-quarterback: "Micah Parsons," Wikipedia: https://en.wikipedia.org/wiki/Micah_Parsons ; "T. J. Watt," Wikipedia: https://en.wikipedia.org/wiki/T._J._Watt ; "Myles Garrett," Wikipedia: https://en.wikipedia.org/wiki/Myles_Garrett . Garrett's 23 sacks in 2025, the single-season record: NFL.com, <https://www.nfl.com/news/browns-de-myles-garrett-reaches-23-sacks-sets-new-nfl-single-season-record> .

²⁰"Kyle Juszczyk," Wikipedia: https://en.wikipedia.org/wiki/Kyle_Juszczyk (signed by San Francisco in March 2017 to a four-year contract worth about \$21 million, by far the largest for a fullback at the time). Patrick Ricard (Ravens fullback and tight end): https://en.wikipedia.org/wiki/Patrick_Ricard . Two-back (21 plus 22) personnel share by season: course fact sheet \$10, nflverse participation.

²¹"Lamar Jackson," Wikipedia: https://en.wikipedia.org/wiki/Lamar_Jackson (32nd overall pick in 2018; 2019: 1,206 rushing yards, then the single-season record for a quarterback, and 36 touchdown passes; unanimous AP MVP for 2019; MVP again for 2023); "2019 Baltimore Ravens season," Wikipedia: https://en.wikipedia.org/wiki/2019_Baltimore_Ravens_season (14-2; 3,296 rushing yards, the NFL single-season team record, breaking the 1978 Patriots' 3,165; offensive coordinator Greg Roman). Harbaugh's firing and the 2026 Ravens staff: course fact sheet \$9.

²²Josh Allen, 2024 AP MVP: NFL.com, list of 2024 NFL Honors winners. Allen's 76 career rushing touchdowns by the end of 2025, the record for a quarterback: "2025 NFL season," Wikipedia: https://en.wikipedia.org/wiki/2025_NFL_season (course fact sheet rates this LIKELY). Jayden Daniels, 2024 Offensive Rookie of the Year, about 890 rushing yards: "Jayden Daniels," Wikipedia: https://en.wikipedia.org/wiki/Jayden_Daniels ; Washington lost the NFC Championship Game 55-23 at Philadelphia (nflverse schedules).

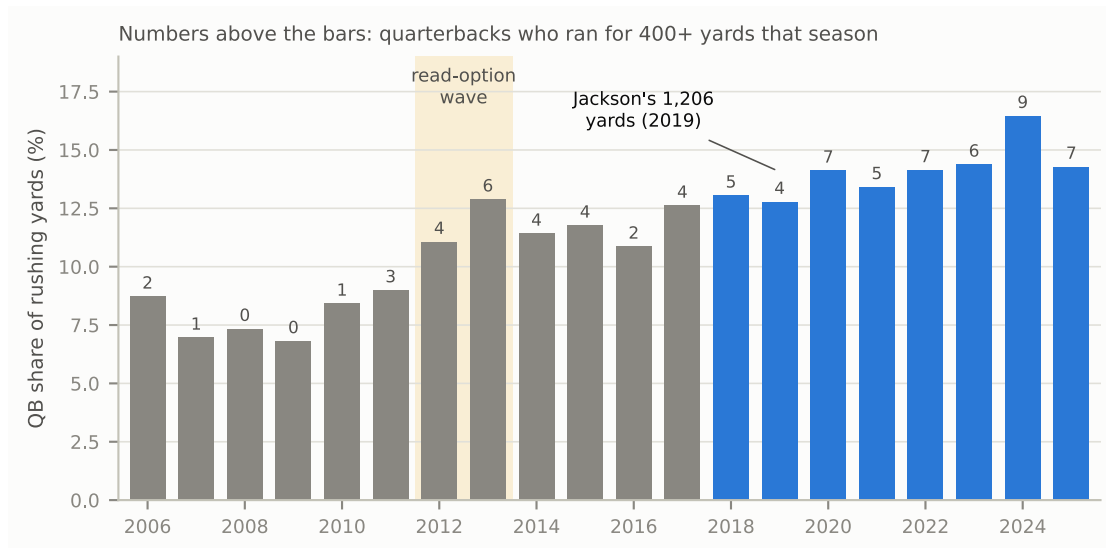


Figure 8: Quarterbacks' share of their teams' rushing yards, 2006-2025: every run by a team's quarterbacks (players with 150 or more pass attempts that season), including designed runs, scrambles and sneaks, as a share of all rushing yards, league-wide, in the regular season. The 2012 read-option wave (shaded) shows as a step up; the Lamar Jackson era is a second, larger step, and in 2024 quarterbacks gained about one rushing yard in six. The numbers above the bars count quarterbacks who ran for 400 yards or more. Source: nflverse play-by-play.

The chart makes two points.²³ First, quarterbacks went from about 7 percent of their teams' rushing yards in the late 2000s to about 14 to 16 percent in 2020-2025, and the number of quarterbacks running for 400 yards in a season roughly tripled. Second, and less obviously, the rise is not a spike and fall like 2012. It is a plateau. Designed quarterback runs, the planned keeps in the read-option chart of the last chapter, actually peaked league-wide around 2020-2022 and drifted down after; what stayed is the quarterback as a runner of every kind: scrambling when the coverage is good, sneaking on third-and-1, keeping on the read when the edge chases. A running quarterback stopped being a scheme and became a skill that most good offenses expect their quarterback to have.

Why did it stick this time, when the 2012 wave faded in two seasons?

- **The arithmetic got better for it.** A quarterback who is a real running threat is an extra blocker's worth of math, the "plus one" of Option Football: the defense has to account for him, so one defender who would otherwise chase the back has to stay home. Against a two-high defense with six in the box, that turns six-against-six into a numbers advantage without changing personnel. The two-high counterrevolution made the running quarterback more valuable, not less.
- **The quarterbacks were built for it, and the offenses were built around them.** The 2012 wave grafted college plays onto NFL offenses for a season. Jackson's Ravens, Hurts's Eagles and Allen's Bills designed their whole run game, protection rules and passing game around a quarterback who ran, and drafted and paid him to be that player for a decade.
- **They learned to run like quarterbacks.** The modern running quarterback slides, gets out of bounds, and picks his moments; a large share of his yards come on scrambles, when the defense is in coverage and nobody is waiting to hit him. The read-option quarterbacks of 2012 took far more hits for every yard.
- **The short-yardage sneak became nearly automatic.** The tush push turned the quarterback's legs into the league's most reliable fourth-and-1 play. The league voted on banning it in May 2025: the proposal failed 22 to 10, two votes short. In 2026 no team even proposed a ban.²⁴

²³This book's calculation from nflverse play-by-play, regular seasons 2006-2025: all run plays by each team's quarterbacks (players with 150 or more pass attempts that season), including scrambles and sneaks but not kneel-downs, as a share of all rushing yards. 7.0 percent (2007), 6.8 (2009), 11.1 (2012), 12.8 (2019), 14.1 (2020), 16.5 (2024), 14.3 (2025). Quarterbacks with 400 or more rushing yards: 1 in 2007, 4 in 2012, 4 in 2019, 9 in 2024, 7 in 2025. Designed quarterback runs per 100 plays (the measure in *Dynasties* and *Counterpunches*): 1.39 in 2020 and 2022, 0.89 in 2025.

²⁴Green Bay's proposal to ban pushing or pulling the runner failed 22-10 on May 21, 2025, two votes short of the 24 required; no ban was proposed for 2026, and the play is legal this season: SI, <https://www.si.com/nfl/packers/packers-proposal-to-ban-eagles-tush-push-fails-again> ; NFL.com, <https://www.nfl.com/news/competition-committee-co-chairman-rich-mckay-doesn-t-expect-tush-push-ban-proposal-in-2026> .

The 2019 Ravens went 14-2 and led the league in points. Watch them for what the opponents had to do. Every running play has two possible ball carriers, so the edge defender on the side away from the back can't chase him; he has to stay home for Jackson. Every play-action pass carries the threat that Jackson will run it himself, so the defense's last line, the safeties, have to think about the run longer than they want to. The Ravens added to the problem with personnel: they used two tight ends, and often a fullback and three tight ends, far more than the league did.

What to notice: count the defenders who are watching Jackson rather than the back at the mesh, the moment the ball is in both their hands. If it's one, the Ravens have the numbers somewhere else. The full case study is The Ravens' Option Run Game, which treats 2019-2025 as a closed era: John Harbaugh was fired in January 2026 and is now the Giants' head coach.

The coaches start trusting the math

The last chapter ended with an argument: David Romer's 2006 paper and Brian Burke's public models said coaches kicked on fourth down far too often, and coaches mostly ignored them. In this era they stopped ignoring them, and you can watch the moment it happened.

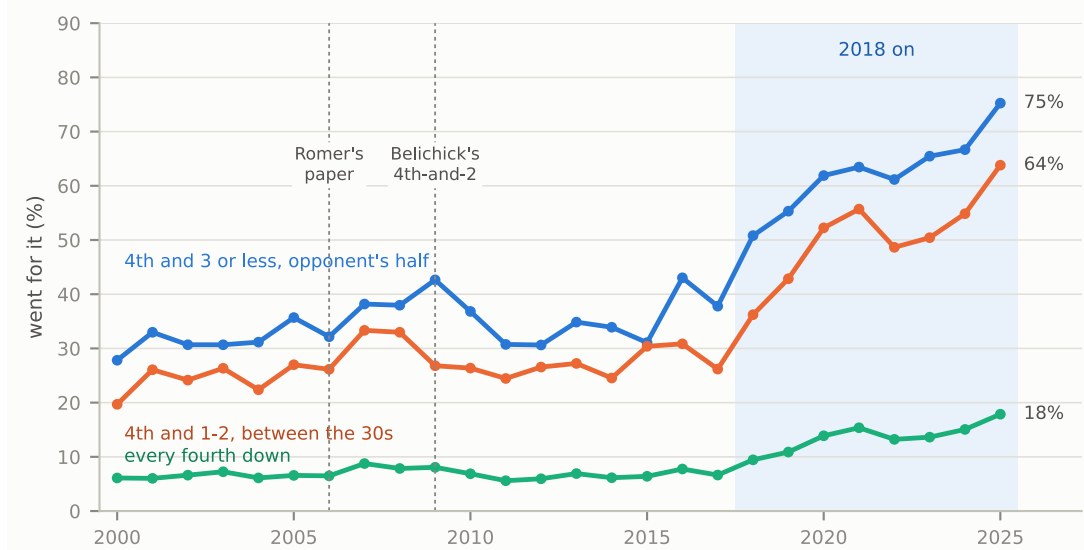


Figure 9: How often NFL teams went for it on fourth down, 2000-2025, in competitive situations (the offense's win probability between 20 and 80 percent, more than two minutes left in the half; a 'go' is a run or pass rather than a punt or field-goal try). Three cuts: fourth-and-1 or 2 between the 30-yard lines; fourth-and-3 or less in the opponent's half; and every fourth down. For eighteen seasons nothing moved, through Romer's paper (2006) and Belichick's famous fourth-and-2 (2009). From 2018 every line roughly doubles. Source: nflverse play-by-play.

The flat stretch is the striking part.²⁵ From 2000 to 2017, on fourth-and-1 or 2 between the 30s in a close game, teams went for it about a quarter of the time, in almost every season. Romer's paper didn't move it; Belichick's 2009 decision, and the uproar after it, didn't move it. Then, in five seasons, the rate doubled, and by 2025 teams went on about 64 percent of those downs and on about three-quarters of short fourth downs in the opponent's half. Why then?

- **The models went public, and onto the broadcast.** Ben Baldwin and Sebastian Carl's nflfastR, released in 2020, put expected points and win probability for every play since 1999 in anyone's hands, free (it is the ancestor of the nflverse data behind every chart in this book). Fourth-down calculators appeared on websites and then on broadcasts, and a coach who punted on fourth-and-1 at midfield heard about it the next morning from people with numbers.

²⁵This book's calculation from nflverse play-by-play, regular seasons 2000-2025, reproducible from the code above the figure: fourth downs with the offense's win probability between 20 and 80 percent and more than two minutes left in the half, "go" meaning a run or pass. Fourth-and-1 or 2 between the 30s: 20-33 percent every season 2000-2017, 36 (2018), 52 (2020), 64 percent (2025). Fourth-and-3 or less in the opponent's half: 28 (2000), 38 (2017), 51 (2018), 75 percent (2025). All fourth downs: 6-9 percent 2000-2017, 18 percent in 2025. Most go-for-its in those situations: Detroit in 2021 (18), 2022 (20) and 2023 (19); the 2021 Chargers were second (17).

- **Teams hired analysts, and listened to them.** By the early 2020s nearly every team had an analytics staff, and several head coaches talked openly about using a fourth-down chart on game day.
- **A champion did it on the biggest stage.** Doug Pederson’s 2017 Eagles were among the most aggressive teams in the league on fourth down, and on fourth-and-goal from the 1 at the end of the first half of Super Bowl LII they ran the “Philly Special,” a direct snap to the running back, a pitch to the tight end and a touchdown pass to the quarterback, Nick Foles. They beat New England 41-33.²⁶
- **New coaches with new habits.** The next wave of head coaches (Brandon Staley’s 2021 Chargers, Dan Campbell’s Lions, Nick Sirianni’s Eagles) went for it constantly. In competitive situations, by this book’s count, Campbell’s Lions went for it more than any other team in 2021, 2022 and 2023.

i Film room: Lions at 49ers, NFC Championship Game, January 28, 2024

The decision that tested the new orthodoxy. Detroit led San Francisco 24-7 at halftime. Early in the third quarter, leading 24-10, the Lions faced fourth-and-2 at the 49ers’ 28, inside field-goal range, and went for it; Jared Goff’s pass fell incomplete. Late in the fourth quarter, now trailing 27-24, they faced fourth-and-3 at the 49ers’ 30, where a field goal would have tied the game, and went for it again; another incompleteness. San Francisco won 34-31.²⁷

What to notice: the same thing 2009’s fourth-and-2 taught. Both decisions were judged by their outcomes the next morning, and both were defensible on the numbers. (Models differed on how close each call was; none treated going for it as the reckless choice it was called.) What the night really showed is how much the culture had changed: in 2009 Belichick’s go was a scandal because coaches didn’t do that. In 2024 Campbell’s were a scandal because they failed. Nobody argued that a coach shouldn’t go for it on fourth-and-2 at the opponent’s 28; they argued about this one. Build the model yourself in Win Probability and Fourth Downs.

Aggression spread beyond fourth down. Teams passed more on early downs than the situation predicted (Tendencies, PROE, and Charting measures that), went for two more often, and used the new overtime rules (Clock and Game Management) to plan whole possessions around a coin flip. The common thread is the go-for-it decision’s logic: a decision is right or wrong before the snap, on its probabilities, not after it, on its result.

The rules underneath, 2017-2026

The rules kept moving too. Most of the era’s changes were about safety, and some of those changed strategy more than any scheme did. One table, with the chapter that owns each rule:²⁸

Season	Rule	What it changed
2017	Regular-season overtime cut from 15 to 10 minutes	More ties possible; the first possession matters more (Clock and Game Management)
2018	Use of the helmet: lowering the head to initiate contact with the helmet is a foul, for any player	Tackling technique: defenders taught to tackle with the shoulder and head up (Contact Rules and Penalties)

²⁶“Philly Special,” Wikipedia: https://en.wikipedia.org/wiki/Philly_Special (fourth-and-goal from the 1 with 38 seconds left in the first half of Super Bowl LII, February 4, 2018; direct snap to Corey Clement, pitch to Trey Burton, touchdown pass to Nick Foles); “Super Bowl LII,” Wikipedia: https://en.wikipedia.org/wiki/Super_Bowl_LII (Eagles 41, Patriots 33). On Pederson’s 2017 Eagles and fourth downs: nflfastR: Ben Baldwin and Sebastian Carl, <https://www.nflfastR.com/>.

²⁷nflverse play-by-play of the 2023 season’s NFC Championship Game (January 28, 2024, Detroit at San Francisco): Detroit led 24-7 at halftime; third quarter, 7:03, fourth-and-2 at the San Francisco 28 with Detroit leading 24-10, Goff incomplete; fourth quarter, 7:38, fourth-and-3 at the San Francisco 30 with San Francisco leading 27-24, Goff incomplete; final 34-31.

²⁸Sources, row by row: overtime (2017 regular season to 10 minutes; 2022 playoffs and 2025 regular season, both teams possess): NFL Football Operations, <https://operations.nfl.com/rules-officiating/featured-rules> and the course fact sheet §4; use of the helmet (2018): Contact Rules and Penalties as Strategy and its sources; 2018 kickoff alignment changes: Kickoffs and Returns; pass interference review, adopted for 2019 after the January 20, 2019 NFC Championship Game and not renewed for 2020: “Pass interference,” Wikipedia: https://en.wikipedia.org/wiki/Pass_interference; dynamic kickoff (2024 trial; permanent in 2025 with the touchback at the 35; return rates about 22 percent in 2023 and 75 percent in 2025): NFL.com, <https://www.nfl.com/news/nfl-owners-vote-to-make-dynamic-kickoff-permanent-adjust-ball-spot-on-touchbacks-to-35-yard-line> and the course fact sheet §3; hip-drop tackle ban (March 2024): NFL.com, <https://www.nfl.com/news/nfl-owners-vote-to-ban-hip-drop-tackle-at-annual-league-meeting>; third challenge (2024): NFL Football Operations, <https://operations.nfl.com/updates/the-rules/approved-2024-playing-rules/>; replay assist (2025) and the 2026 changes: the course fact sheet §6 and the sources there.

Season	Rule	What it changed
2018	Kickoff alignment rules (no running start; five players on each side of the kicker)	The first of many kickoff redesigns (Kickoffs and Returns)
2019	Pass interference reviewable by replay, for one season only	After the Rams-Saints NFC Championship no-call; almost never overturned, then dropped
2022	Playoff overtime: both teams get a possession	After the Bills-Chiefs “13 seconds” game; changes two-point and kick-or-receive math
2024	The dynamic kickoff (a one-year trial); hip-drop tackle banned; a third challenge if one of the first two succeeds	Kickoff returns come back; a tackling technique outlawed
2025	Dynamic kickoff made permanent, touchback to the 35; regular-season overtime gives both teams a possession; replay assist can correct some fouls; tush-push ban fails 22-10	Kickoff returns on about three-quarters of kicks; field position shifts toward the offense
2026	Onside kick may be declared at any time, even when leading; a kickoff from the 50 that goes out of bounds is a touchback at the 20; a fourth “floater” in the receiving team’s setup zone; replay officials may be consulted on ejections	Changed after the 2025 season: the newest kickoff tweaks (Kickoffs and Returns)

Two patterns in the table matter for how you watch. First, **kickoffs changed more than any other part of the game**: four redesigns in nine seasons, and by 2025 a kickoff was returned about three times out of four, where in 2023 it was returned about one time in five. Starting field position, and so the fourth-down and two-minute math, moved with it. Second, **the league kept protecting the passing game indirectly**. The helmet rule, the hip-drop ban and replay assist are safety rules, but every one of them makes defenders a little more careful, and the 2022 and 2025 overtime changes reward offenses that can score touchdowns, not field goals. Nothing in nine seasons tilted the field back toward defense; the defenses had to do that with scheme.

Where the game stands entering 2026

Which brings us back to Seattle.

Mike Macdonald built his reputation as the Ravens' defensive coordinator in 2022 and 2023, when Baltimore allowed the fewest points in the league and led it in sacks. His defenses start from two-high shells and then disguise everything after: safeties rotating at the snap, simulated pressures that show six rushers and send four (Zone Blitz, Fire Zones, and Simulated Pressure), and linebackers standing in the A gaps before the snap. He became Seattle's head coach in 2024.²⁹

In 2025 he hired Klint Kubiak, Gary Kubiak's son and a Shanahan-tree coach, as offensive coordinator, and the Seahawks played the other half of this chapter: under center on 53 percent of their snaps, fewer than three wide receivers on 55 percent, a fullback on the field more than most teams, outside zone and play-action. They went 14-3, allowed the fewest points in the league, and beat New England 29-13 in Super Bowl LX, with Jason Myers kicking a Super Bowl-record five field goals, Uchenna Nwosu returning an interception for a touchdown, and Walker running for 135 yards. The defense played two-high on about 52 percent of opponents' dropbacks, the third-highest rate in the league, and still sacked Drake Maye six times.³⁰

What to notice: that the same team is the era's answer on both sides. On defense, a two-high shell that hides pressure, so the light box isn't a gift. On offense, a heavy, under-center run game built to beat exactly that kind of defense. When you watch the 2026 Seahawks, watch whether the offense keeps its shape without the coordinator who built it: Kubiak left after one season to become the Raiders' head coach, and Brian Fleury, from Shanahan's staff, replaced him.

The coaching map that Seattle sits on is changing fast. Kubiak to Las Vegas; Mike McDaniel, fired in Miami, now the Chargers' offensive coordinator; Mike LaFleur, McVay's 2025 coordinator, now Arizona's head coach; John Harbaugh in New York; a fifth straight new offensive coordinator in Philadelphia. The Shanahan tree, which in 2025 supplied play-callers all over the league, is scattering into new branches, and the 2026 staffs are not the 2025 ones.³¹ Label every staff with its season when you read about these teams.

And the rules enter 2026 changed again: onside kicks any time, a kick from the 50 out of bounds now a touchback at the 20, a fourth floater in the return team's setup, replay allowed to advise on ejections, the tush push still legal, and no 18-game season yet.³²

The pattern: an arms race you can watch

Step back and the nine seasons make a loop, the same rhythm the last chapter found, only faster.

²⁹"Mike Macdonald (American football)," Wikipedia: [https://en.wikipedia.org/wiki/Mike_Macdonald_\(American_football\)](https://en.wikipedia.org/wiki/Mike_Macdonald_(American_football)) (Ravens defensive coordinator 2022-2023; Seahawks head coach from 2024). The 2023 Ravens allowed 280 points, fewest in the league, and led it with 60 sacks: this book's calculation from nflverse schedules and play-by-play.

³⁰Seattle's 2025 record (14-3, NFC West champion and No. 1 seed): course fact sheet §2; fewest points allowed (292), and the under-center, personnel and coverage rates: this book's calculation from nflverse data (see the first footnote of this chapter). Super Bowl details (Myers's five field goals, a Super Bowl record; Nwosu's 45-yard interception return for a touchdown, rated LIKELY): course fact sheet §1. Kubiak's single season as offensive coordinator and his move to the Raiders; Fleury as 2026 offensive coordinator: course fact sheet §0 (C2) and §9.

³¹Course fact sheet §9, "Coaching landscape for the 2026 season": Klint Kubiak, Raiders head coach; Mike McDaniel, fired by Miami in January 2026, Chargers offensive coordinator; Mike LaFleur, Cardinals head coach; John Harbaugh, Giants head coach; Sean Mannion, the Eagles' fifth new offensive coordinator in five seasons. Sources listed there, including <https://www.profootballrumors.com/2026/02/2026-nfl-offensive-defensive-coordinator-search-tracker>.

³²Approved March 31, 2026: Seahawks.com, "NFL announces approved rule changes for 2026 season": <https://www.seahawks.com/news/nfl-announces-approved-rule-changes-for-2026-season>; Football Zebras: <https://www.footballzebras.com/2026/04/nfl-owners-approve-rule-changes-for-the-2026-season/>. No 18-game season: Pro Football Rumors (June 2026): <https://www.profootballrumors.com/2026/06/formal-negotiations-on-18-game-season-yet-to-begin>.

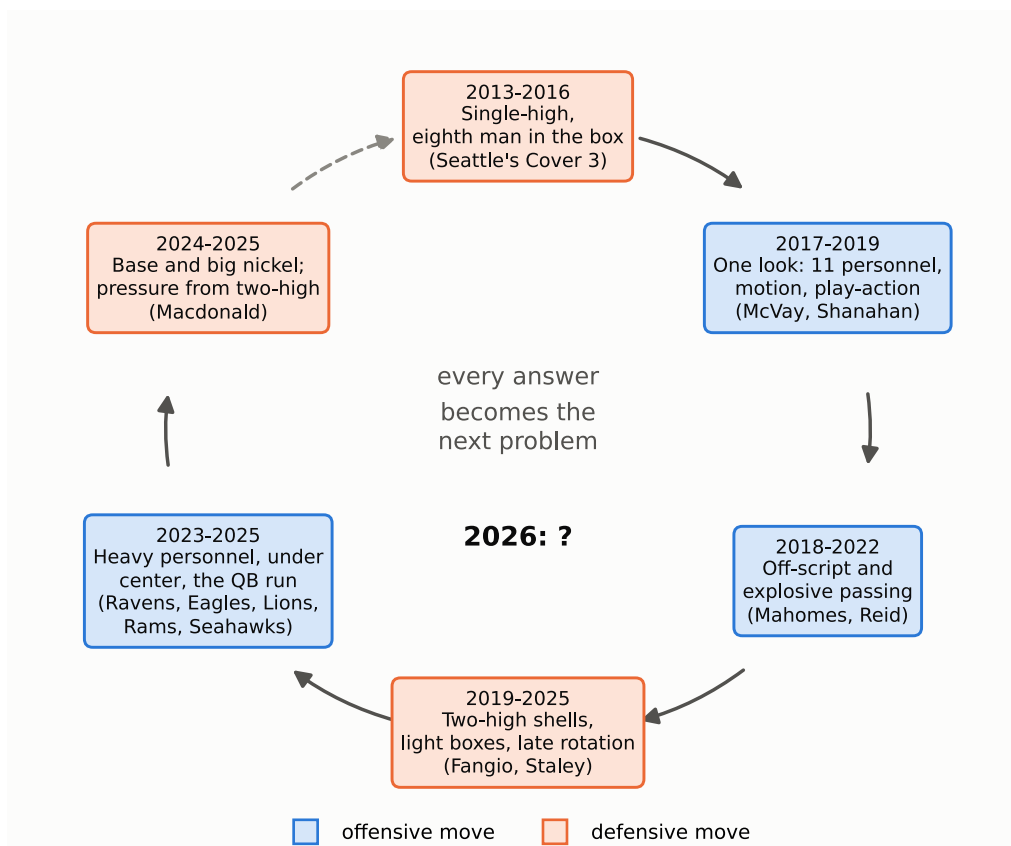


Figure 10: The arms race of 2010-2025 as a loop. Each box is a move; each arrow is the answer it provoked. Start at the top: Seattle’s single-high defense put an eighth man in the box; McVay and Shanahan’s play-action punished him; Mahomes punished anyone left alone deep; two-high defenses took away both and emptied the box; offenses answered with heavier personnel, the quarterback run and the snap from under center; and defenses answered that with base and big-nickel packages and pressure disguised inside a two-high shell. The dashed arrow is the open question entering 2026: what beats that?

The loop is the best single tool this chapter can give you, because it tells you what to look for next. Each box was a sensible answer to the box before it, and each one created a weakness the next box attacked. The defense of 2025 (two-high, but with base personnel and disguised pressure ready for heavy offenses) is an answer to heavy offenses. So the obvious question for 2026 is the one the last chapter already asked: what does a heavy offense do when the defense matches it with big players? It spreads them out. A team that lines up in 12 personnel against a base defense and then splits its tight ends wide is asking a linebacker to cover a receiver, the Patriots’ dilemma of 2011, fourteen years later. Whether that, or something nobody has thought of, is the next arrow is what the 2026 season will tell you.

The move	The answer	What it opened up
Single-high, eighth man in the box (Seattle)	McVay-Shanahan: one look, motion, play-action into the middle	Defenses needed to stop committing to the run before the snap
Off-script, explosive passing (Mahomes)	Keep everything in front: more two-high	Light boxes
Two-high, light boxes (Fangio, Staley)	Light-box run game: heavy personnel, under center, the running quarterback	Safeties forced to fit the run, and play-action behind them
Heavy personnel and the QB run	Base and big nickel; pressure disguised in a two-high shell (Macdonald)	Big defenders who can be spread out and isolated
Kicking on fourth down	Going for it, on the models’ advice	Every fourth-and-short is now a play-call

💡 Watch for it on Sunday: arms-race spotting

Pick one game and find one link of the loop in it: one offensive trend, and the defensive counter you can see against it.

- **Before the first snap, set your baseline.** What personnel does the offense open with, and is the quarterback under center or in the shotgun? Count the deep safeties: one or two?
- **On first and second down, count the box.** Six or fewer with two safeties deep is a two-high, light-box defense. Is the offense running into it, and is the run getting five yards?
- **If the run is working, watch the safeties.** Does one start walking down before the snap, or sprinting down at the snap? That is the defense giving up its shell. Now watch for play-action behind him.
- **Count the tight ends and backs.** Two tight ends or a fullback on first down is the heavy answer. Does the defense match with a third linebacker or a third safety? If it does, does the offense then split a tight end out wide against a linebacker?
- **Watch the quarterback when the pocket breaks.** Does he run, and does the defense have a spy or a rusher who stays home for him?
- **On fourth-and-short in the opponent's half, predict before the coach decides.** In 2025 teams went for it about three times in four in that situation. Did this coach?
- **By halftime, name the link.** "The offense is running heavy into two-high boxes, and the defense just started rotating a safety down" is one. Write it down and see whether it changes in the second half.

Predict the play

💡 Predict the play 1: the safety who keeps creeping down (2019)

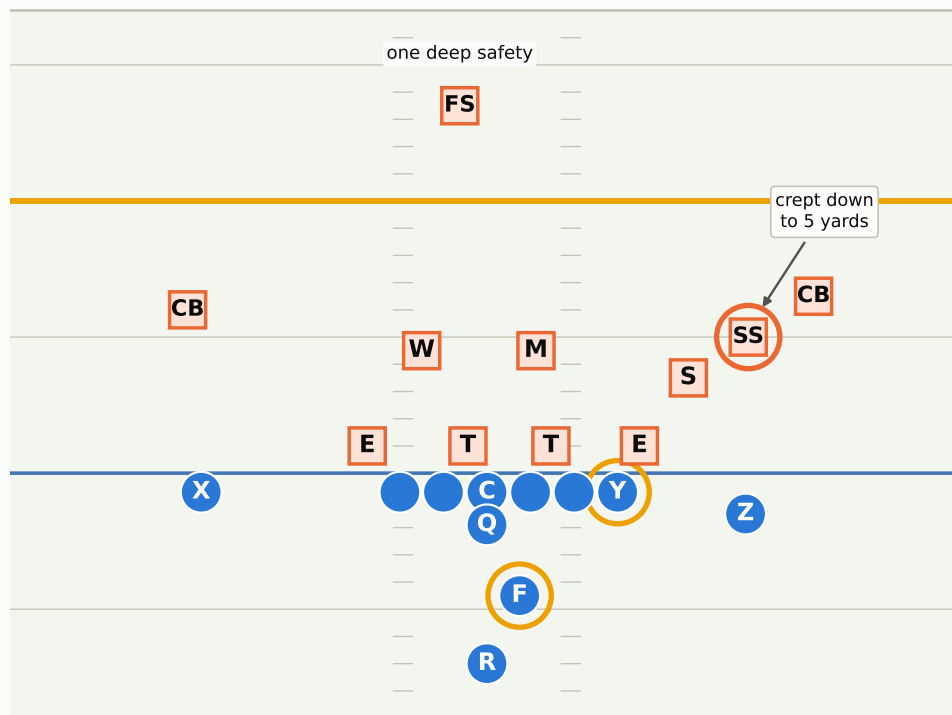


Figure 11: 2019, first-and-10 at midfield. A Shanahan-style offense in 21 personnel (fullback F and tight end Y, gold rings) under center; the receivers are in tight splits. It has run outside zone to the right three times this quarter for 5, 7 and 6 yards. The defense is single-high, and the strong safety (SS, ringed) has crept down to 5 yards, outside the tight end: eight defenders are in the box. The two linebackers are at 4.5 yards, eyes on the backfield.

The offense shows the same look a fourth time. What's the call, and where does the ball go?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: second-and-6 against two-high (2022)

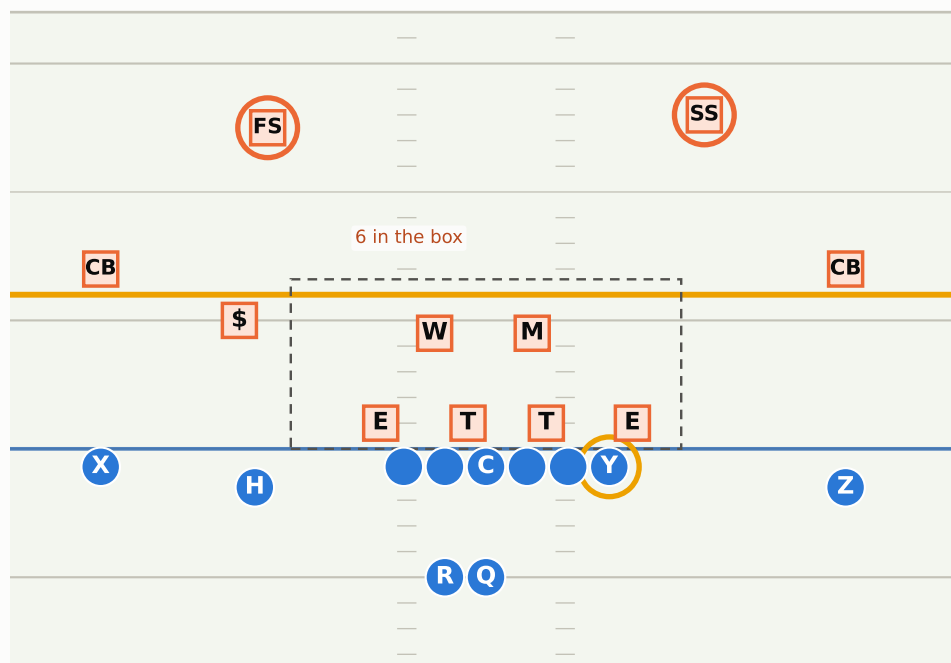


Figure 12: 2022, second-and-6. 11 personnel in the shotgun, tight end (Y, gold ring) attached on the right, a 2x2 receiver set. The defense is in nickel with both safeties (ringed) 12 to 13 yards deep: four linemen and two linebackers in the box (dashed rectangle), the nickel back (\$) out over the slot receiver, both corners off.

Count the box and the blockers. Run or pass, and why is the defense comfortable with your answer?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: heavy personnel, no huddle (2025)

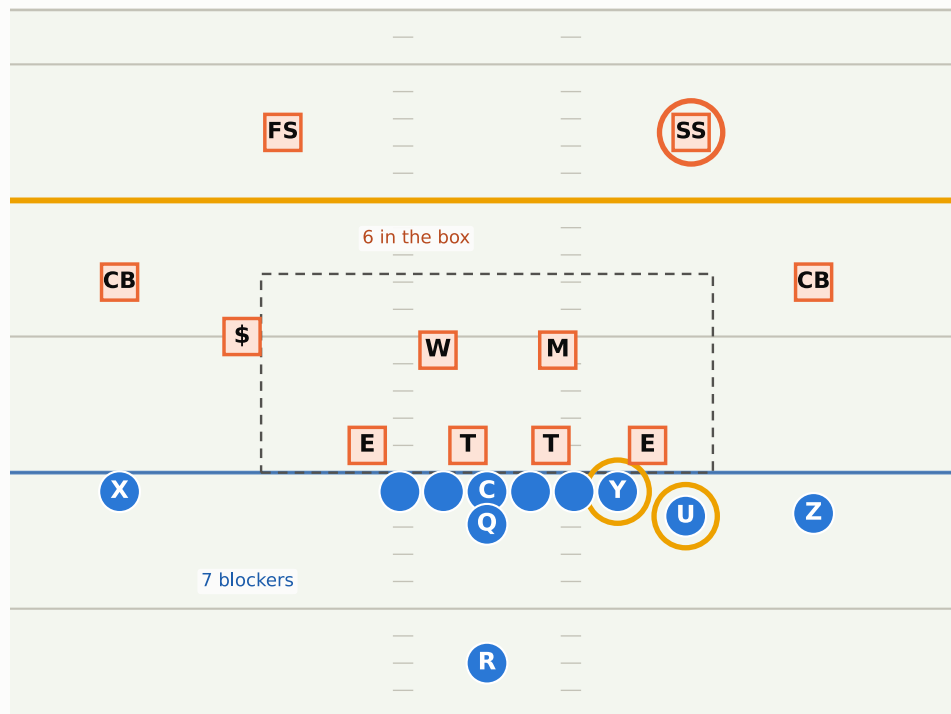


Figure 13: 2025, first-and-10. On the previous play the offense used three receivers and the defense matched with nickel. Now, without huddling, the offense has lined up in 12 personnel (two tight ends, gold rings) with the quarterback under center, and the defense, unable to substitute, is still in nickel with two safeties deep. The strong safety (SS, ringed) is the defender whose job, in a light box, is to come down and fit the run.

You're the defensive coordinator, and you can't substitute. What are your options on this snap, and which one is the offense hoping you pick?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **McVay and Shanahan won 2017-2019 with sameness.** One personnel group, tight splits, motion, and play-action off the outside zone from under center made every play look like a run, and pulled the linebackers and the box safety of single-high defenses out of the intermediate middle.
- **Mahomes and Reid added what a defense can't scout:** a college-borrowed passing game and a quarterback whose best plays came after the design broke down. Together with McVay, they pushed defenses to keep more players deep.
- **The two-high counterrevolution** (Fangio, Staley, and their trees) kept two safeties deep, played the run with six in the box and safeties fitting late, and disguised its coverage until the snap. Two-high coverage rose from about a third of dropbacks in 2018 to a record 42 percent in 2025; the price was light boxes and four- to five-yard runs.
- **Offenses answered with the light-box run game and the under-center revival.** More blockers than the defense has box defenders (12 and 13 personnel, the fullback, the running quarterback), a run from under center that looks exactly like the play-action pass, and a deep safety forced to guess. In 2025 offenses used fewer than three receivers on about 42 percent of plays and went under center on about a third.
- **Positions mutated to fit the race:** the nickel became the base, the star and big nickel answered heavy sets, linebackers got lighter, the move tight end became the offense's hinge, the edge rusher became the premium defender, and the fullback found a niche. And the **quarterback run became normal**, because it adds a blocker's worth of math against a light box and because the quarterbacks were built and protected for it.

- **Coaches started trusting the math, and the rules kept moving.** Fourth-and-short go rates doubled after 2018; kickoffs, overtime and contact rules changed almost every year. Seattle’s Super Bowl LX team, two-high on defense and heavy and under center on offense, is where the race stood entering 2026.

Connections

Builds on: Dynasties and Counterpunches (the move tight end, the hybrid edge, the 2012 read-option wave and the first analysts), Disguise, Rotation, and Split-Field Coverage (two-high shells and rotation), Run Fits and Box Math (the light box and how a defense fits the run with safeties), Play-Action, Boots, and Screens and Motion and Shifts as Weapons (the McVay-Shanahan toolkit), and Personnel Groupings (the substitution game).

Sets up: the Part 12 case studies, each of which takes one thread of this chapter and follows it all the way: The Shanahan Tree, Andy Reid and the Kansas City Chiefs, The Ravens’ Option Run Game, The Eagles, The Lions and Fangio’s Two-High Revolution and Macdonald’s Answer. For the numbers, Win Probability and Fourth Downs and Tendencies, PROE, and Charting; for the money behind the edge rusher and the nickel, Building a Roster. The rules timeline in Appendix E has every rule change in one list.

If you want to go deeper: Seth Walder’s and Ben Solak’s work at ESPN and The Ringer on motion, play-action and fourth downs; the NFL’s Next Gen Stats articles (James Reber’s “trend watch” series has the two-high and heavy-personnel numbers); Cody Alexander’s *MatchQuarters* writing and books for the defensive side of the two-high era; Ted Nguyen’s film breakdowns of the Shanahan and McVay offenses; and, for the data, the nflverse project itself, with Ben Baldwin’s open-source tutorials on how to rebuild every chart in this chapter.

Answers to Predict the play

Answer 1.

The play-action pass off the same outside-zone look, and the ball goes where the creeping safety and the linebackers used to be. Three straight runs of 5 or more yards are what made the safety creep down; he is now a ninth run defender, and the defense has one deep safety for the whole field. Show the fourth outside zone (the line steps right, the back and fullback run the same path), and the linebackers and the strong safety will step toward it, because the last three times they were right to.

The most likely call in this offense is a boot away from the fake: the quarterback fakes right and rolls left, the fullback slides into the left flat, X runs a vertical to take the left corner deep, and Z, from the right, runs the deep crossing route across the field behind the linebackers, as in the play-action figure earlier in the chapter. Against a defense this committed, the tight end leaking up the seam behind the strong safety is the shot. Whatever the route, the principle is the McVay-Shanahan one: the run earned the pass.

And notice what this picture taught defenses. A coordinator who keeps losing this play has two answers: stop putting the eighth and ninth men near the line before the snap, or put them there and bring them back at the snap. Both lead to two-high.

Answer 2.

Run. The box has six defenders (four linemen, two linebackers) and the offense has six blockers in front of the back (five linemen and the attached tight end). Every box defender has a man, and the two safeties are at 12 and 13 yards, so a run that gets through the line has five or six yards before anyone can arrive. On second-and-6 that is a first down or a third-and-short. Inside zone or duo straight at the box is the call; a quarterback who is a running threat makes it better still, because a read on the backside end turns six-on-six into a numbers advantage.

Why is the defense comfortable? Because this is the trade it chose. A five-yard run here is worth a little less than nothing to the offense on average (runs against light boxes in this era averaged about -0.04 EPA), and the defense has kept both safeties over the top of every receiver, so the play that would really hurt it, the 25-yard completion, is much harder. It is also not as static as it looks: one of those safeties may be rotating down at the snap, which turns your “light box” into seven and your run into a three-yard gain. That is why the offenses of 2023-2025 stopped settling for six-on-six and added a seventh blocker.

Answer 3.

You have three options, and the offense has a play for each.

- **Stay in two-high with six in the box.** The offense has seven blockers (five linemen and two tight ends, both on the right) against your six box defenders, plus a back. That is an outside-zone run to the tight ends’ side, and your nickel back (\$) is on the far side of the field. Expect five yards or more.

- **Rotate the strong safety down at the snap** to make it seven against seven. Now you're single-high, and the strong safety is fitting a run that may be a fake: this is the play-action figure of the under-center section, the post or the deep over behind the safety who came down. The offense is under center precisely so that its run and its fake look the same to him.
- **Slant and stunt the front, or bring pressure from the two-high shell**, the Macdonald answer: move your linemen at the snap so the blocking angles are wrong, or send a linebacker into the gap the extra blocker was meant for while the safeties stay home. This is a guess, and against the right call it wins big; against the wrong one, the run goes for a lot.

Which does the offense want? Usually the second. The heavy, under-center offense is designed to make the run good enough that the safety must come, and then to throw behind him; that is where its big plays are. That is also why, after this snap, the defense will try very hard to get a base package or a third safety on the field the next time those two tight ends come in, and why the offense will then try to split them out wide against linebackers. The 2025 rise in base defense, from 20.5 percent of snaps in 2023 to 29.7 percent, is that next move.